



**Performance Work Statement
for
Product Data Systems**

**Air Force Life Cycle Management Center
Technical Services Division
AFLCMC/LZP-Robins**

05 JANUARY 2026



TABLE OF CONTENTS

1. DESCRIPTION OF SERVICES.....	5
1.1. Objectives	5
1.2. Benefit to United States Air Force.....	5
1.3. Background.....	5
2. SERVICES SUMMARY	6
3. GOVERNMENT FURNISHED PROPERTY (GFP) / GOVERNMENT FURNISHED MATERIAL (GFM) AND SERVICES.....	7
3.1. Government Inventory	7
3.1.1. Government Inventory Reports	7
3.1.2. Government Furnished Property (Material/Equipment).....	7
3.1.3. Contractor Responsibility	8
3.2. Government Property to be Provided	8
3.3. Base Support	8
3.4. Reporting of Government Property	9
3.5. Loss of Government Property	9
3.5.1. Definition of Loss of Government Property.....	9
3.5.2. Reporting Loss of Government Property	9
3.6. Return or Retention of Government Property.....	9
4. TECHNICAL REQUIREMENTS	9
4.1. Period and Place of Performance	9
4.1.1. Contract/Order Period of Performance.....	9
4.1.2. Hours of Operation	10
4.1.3. Accommodation.....	10
4.1.4. Location	11
4.1.5. Travel.....	11
4.1.6. Remote Work.....	11
4.2. Contract Data Requirements List(s) (CDRL)	12
4.2.1. Quality of CDRL Deliverable.....	12
4.2.2. Receipt of CDRL Deliverable	12
4.3. Antiterrorism (AT).....	13
5. TECHNICAL SUPPORT	13
5.1. Technology Support.....	13
5.2. System Support Services.....	13
5.3. System Sustainment	14
5.3.1. Help Desk Sustainment	15

5.3.2. Operations Sustainment.....	15
5.3.3. Cybersecurity Sustainment.....	16
5.3.4. Development sustainment.....	17
5.4. Software Licensing/Testing	18
5.5. ‘Warm’ Site and System Backup Capabilities.....	18
5.6. System User Training	19
5.7. Production Publishing Support	19
5.8. Content Management System Implementation/Support	20
5.9. S1000D Specification Data Capabilities.....	21
5.10. Subject Matter Expert Support.....	22
5.10.1. Transformation and Migration Support Services	23
5.10.2. Enterprise Level Capabilities/Data Support	30
5.10.3. Documentation.....	32
5.10.4. Project Management/Reports	34
5.10.5. Data Rights	35
5.10.6. Support Meetings/Conferences.....	35
5.10.7. Reserved	35
5.10.8. Reserved	35
5.10.9. Reserved	35
5.10.10. Required Skills	35
6. GENERAL INFORMATION: PRODUCT DATA SYSTEMS SUPPORT	37
6.1. Security Requirements	37
6.1.1. Security Regulations.....	37
6.1.2. Operations Security (OPSEC)	37
6.1.3. Communications Security (COMSEC)	37
6.1.4. Security Clearance	37
6.1.5. Security Incident or Violation	38
6.1.6. Access to Government Systems	38
6.1.7. Access to Robins AFB.....	39
6.1.8. Contractor Identification.....	39
6.1.9. Common Access Card (CAC)	39
6.2. Environmental Management (EMS)	40
6.3. Agency Affirmative Procurement Programs	40
6.4. Safety Requirements	40
6.4.1. Mishap Notification/Investigation.....	40

6.4.2. Safety and Health Plan/Program.....	41
6.5. Inspection of Services	41
6.6. Service Contract Reporting (SCR).....	41
6.7. Invoicing/Payment and Receipt/Acceptance	41
6.8. Quality Control Plan/Quality Management (QMS).....	41
6.8.1. Self-Assessment Program (SAP).....	41
6.8.2. Quality Control and Continuous Process Improvement (CPI)	42
6.9. Trafficking in Person	42
6.10. Continuation of Mission Essential Services During a Crisis	42
6.10.1. Designation of Services as Mission Essential	42
6.10.2. Mission Essential Contractor Services	42
6.11. Nondisclosure Requirements	43
6.12. Personnel Reporting.....	43
6.13. Uses and Safeguarding of Information	44
6.14. User Manuals	44
6.15. Non-Personal Services	44
6.16. Transition/Phase Out.....	44
6.17. Government Points of Contact.....	45
7. APPENDICES	45
7.1. Appendix A. Reference(s)	45
7.2. Appendix B. Product Data Systems Hardware and Application Descriptions	47
8. EXHIBITS	51
8.1. Exhibit Reference(s):	51

DESCRIPTION OF SERVICES

1.1. Objectives

This Performance Work Statement (PWS) describes the services required to support Department of the Air Force (DAF) Product Data Systems Support (PDSS). These services include systems management, data management, and enterprise migration and sustainment support required for continued performance of highly specialized technical services at Robins Air Force Base (AFB) supporting product data systems and infrastructure, data management, conversion and migration processes, and transformation initiatives.

1.2. Benefit to United States Air Force

This acquisition will benefit the DAF by providing processes and tools to enable the enterprise division and its customers to acquire, develop, manage, transform, inspect, store, distribute, and maintain technical data.

1.3. Background

The contractor shall provide all labor, supplies, and technical support services necessary to operate and maintain Robins AFB product data hardware and software; to aid Government customers in managing and executing technical data generation, acceptance and sustainment activities, processes, and products within the product data environment; and to support data improvement, cleansing, and migration efforts necessary to prepare data for transition into modernized and/or DAF enterprise-level systems, such as the Enhanced Technical Information Management System (ETIMS), the Technical Order Authoring and Publishing (TOAP) enterprise, Air Force Product Lifecycle Management (AF-PLM), solutions resulting from DAF Logistics Capability Initiatives (CI), solutions resulting from DAF AFMC/A4 Logistic Information Technology (LogIT) initiatives, solutions resulting from AF-PLM Enterprise solution initiatives, and other system solutions which may result from system and data center consolidation activities.

ETIMS provides the capability for DAF system maintainers to manage their Technical Order (TO) and Electronic TO (eTO) subscriptions and libraries. Much of the data processed and/or managed within the Robins AFB Product Data Systems environments must also reside within the ETIMS in order to satisfy end-user requirements. Portions of the requirements outlined within this PWS are specific to the TO document structure and data loading requirements driven by ETIMS.

TOAP is designated as the organic TO authoring and publishing component of the DAF Standard TO Management System. TOAP provides the DAF enterprise capability to manage, author, and publish TO publication source content data and enforces authoring and publishing compliance outline in DAF policy documents.

AF-PLM is a major initiative designed to establish and implement an Air Force enterprise system which will subsume the majority of the existing Robins Technical Data-Product Data Management (TD-PDM) system and the data contained within those systems. Since the AF-PLM solution will provide capabilities/functionality similar to those currently supplied by the Robins AFB technical data systems, the level of technical systems and data support required is expected to increase in preparation for AF-PLM implementation, and then decrease as capabilities and data incrementally transition to that implementation and/or other enterprise-level systems.

At present, the TD-PDM system contains approximately 70,000 active TO documents/increments, comprising an estimated 50,000,000 pages of various sizes; over 800,000 Computer Aided Design/Computer Aided Manufacturing (CAD/CAM) models and files; over 175,000 part indenture configurations consisting of over 18,300,000 part records; and more than 84,000,000 drawing related datasets in A-TEAM are managed at Robins AFB. Additionally, data owned and managed by DAF organizations at locations other than Robins is actively being migrated for sustainment within the PDSS environment at Robins AFB. Services to support product data (TO, engineering, and other product data) systems infrastructure and data management, sustainment, and transformation activities include:

- Technical services to manage, maintain, and administer all computing hardware and software utilized to support the management, creation, sustainment, storage, and distribution of technical and engineering data.
- Technical support to assist the Government in development and execution of digital data support, sustainment, and transformation strategies and processes.
- Technical support necessary to facilitate evaluation and implementation of hardware and software necessary to support transition of technical data systems functionality and data to recently adopted DAF data standards, to newly implemented local system/data environments, and to DAF enterprise-level systems currently in various conceptual, planning, development, and/or implementation stages.
- Technical support to establish, enhance, and execute processes necessary to discover, establish, and maintain relationships/links between various types of product data (i.e., parts, drawings, TOs).

2. SERVICES SUMMARY

Performance Objectives	PWS Paragraph	Performance Threshold
Maintain tech data system availability to all users	5.3	Maintain system uptime rate of 98.5% or higher monthly per system (TD-PDM, TOAP, STOR). (Based on a 30-day month, 100% system uptime would equate to 720 hours.)
Cybersecurity and Authority-To-Operate (ATO) support	5.3	Submit required RMF documentation updates within specified timeframes: SSPs - within 15 days of system changes Hardware/software inventories - within 15 days of changes.
Review and implement release Secure Technical Implementation Guidelines (STIGS)	5.3	Submit all applicable STIG checklist files with comments for each check and zero Not Reviewed checks within 30 days of release.
Maintain software licensing compliance	5.4	Provide notification of illegal software products within 30 days of identification.
Maintain viable system/data backup	5.5	Ensure execution and validity of backup routines at 100%. Full backups shall be accomplished weekly with daily incremental backups accomplished.

Performance Objectives	PWS Paragraph	Performance Threshold
Maintain contingency capabilities	5.5	Monthly certify 100% synchronization and availability of critical data transferred to the COOP.
Complete delivery of PDF quality assurance data	5.10.1.3	Submitted documents priority load: • Routine: within seven (7) working days of receipt • Urgent and Emergency: within three (3) working days of receipt
Ensure accuracy of processed IPB part data	5.10.1.4	Ensure required data accuracy rate of 98% based on MPL achieved from source IPB content.
Development/sustainment of system, system security, and process documentation	5.10.3	Correct identified deficiencies or required updates within 15 working days. Provide a report of which documents were validated, and which ones were updated. Versioning and change summary shall be captured in a change log on each document.
Government Furnished Equipment (GFE) Hardware Inventory	3.4, 3.6	Maintain accurate hand receipts and update within 5 days of equipment assignment or changes.

3. GOVERNMENT FURNISHED PROPERTY (GFP) / GOVERNMENT FURNISHED MATERIAL (GFM) AND SERVICES

Contractor will track, secure, and account for all GFP/GFM IAW FAR 45 and DFARs 245: The property shall be listed as an attachment of the request for proposal package. The fillable form is located at <https://dodprocurementtoolbox.com/site-pages/gfp-attachments>.

3.1. Government Inventory

3.1.1. Government Inventory Reports

The contractor shall perform an inventory of all Government property to include material and equipment not later than thirty (30) calendar days after contract start. After the initial inventory, the contractor shall inventory quarterly thereafter. Inventories may be required intermittently and shall be submitted as requested by the COR or PM. Quarterly physical inventory reports shall be submitted to the Government Property Administrator (AFSC/PZIE and the PMO). The report shall detail the results of the inventory, any items not accounted for and items that are lost, damaged or destroyed beyond use.

(Exhibit A: CDRL A001, Technical Report-Study/Services, DI-MISC-80508B)

3.1.2. Government Furnished Property (Material/Equipment)

The contractor shall maintain records evidencing receipt, accountability, preventative maintenance, and disposal as defined within the Government property clauses. All furnished Government property shall be included in the attached GFP inventory list spreadsheet.

The contractor shall maintain a record of accountably and allocation for all GFP (Material/Equipment) by PDSS major program including software license allocation and hardware allocation by system down to the application level where possible. The contractor shall be able to supply a detailed report of license allocation, and a report of hardware required to support each PDSS major program.

3.1.3. Contractor Responsibility

The contractor shall make recommendations for purchases, as necessary, whenever existing hardware and/or software serviceability must be maintained, expanded or replaced to meet the requirements.

(Exhibit A: CDRL A001, Technical Report-Study/Services, DI-MISC-80508B)

The contractor will be responsible for gathering requirements (maintenance, support level, versioning, etc.), obtaining quotes and executing purchases, and for physical acceptance of procured items.

The contractor shall also recommend for lease or purchase any equipment, furniture, supplies, office space, and any/all other items or services deemed necessary to fulfill the requirements outlined in this PWS. Recommended items are to be acquired through means deemed most advantageous to and approved by the Government. All software developed under this contract shall be provided with unlimited rights. All equipment acquired, or software acquired, developed or licensed to facilitate processes or meet production requirements shall transfer to the Government at contract completion.

The Government shall have unlimited rights to all non-commercial software developed under this contract. Non-commercial source code for all systems, scripts, software tools, etc., developed under this contract must be thoroughly documented and shall be delivered unencumbered by restrictive markings to the Government at contract completion or within 30 days after receipt of a request from the Government Program Manager.

(Exhibit A: CDRL A003, Software Transition Plan, DI-IPSC-81429A /A004, Computer Program End Item Documentation, DI-IPSC-80590B)

Commercial software licenses shall also transfer to the Government at contract completion, or within 30 days after receipt of a request from the Government Program Manager.

(Exhibit A: CDRL A003, Software Transition Plan, DI-IPSC-81429A)

The Government Program Manager shall approve all development efforts prior to start of development. All custom-developed noncommercial software shall be properly documented using industry or DAF standard techniques.

(Exhibit A: CDRL A004, Computer Program End Item Documentation, DI-IPSC-80590B)

3.2. Government Property to be Provided

The Government will provide the contractor access to facilities and the PDSS hardware and software necessary to support the tasks. With the exception of COTS software and standard office/supply items, all required technical data support materials currently utilized shall be provided by the Government. Therefore, the contractor shall utilize Government property to fulfill the requirements of this contract whenever possible.

3.3. Base Support

The Government will furnish property incidental to the place of performance including office space and equipment, i.e., computer equipment, software, output device(s), phone/phone line, facsimile equipment, reproduction equipment and office supplies.

3.4. Reporting of Government Property

The contractor shall have a system of internal controls to manage (control, use, preserve, protect, repair, and maintain) Government property in its possession in accordance with FAR 52.245-1, and its property management system shall be adequate to satisfy the requirements of this clause. The contractor shall record receipt and return of all Government Furnished Property (GFP), as well as the reporting of shipment, consumption, disposal, or transfer to another contract, in the GFP Module in Procurement Integrated Enterprise Environment (PIEE), <https://piee.eb.mil>, as stipulated by DFARS 252.245-7005. This reporting requirement does not change any liability provisions or other reporting requirements that may exist under the contract. GFP training and guidance is located at the DoD Procurement Toolbox at <https://dodprocurementtoolbox.com/site-pages/gfp-training>.

3.5. Loss of Government Property

3.5.1. Definition of Loss of Government Property

“Loss of Government property” means unintended loss of or damage to Government property including property that cannot be found after a reasonable search, loss due to inadequate storage, loss due to lack of security, theft, damage requiring repair to restore the item to usable condition, or damage that renders the property useless for its intended purpose or Beyond Economical Repair (BER). Loss of Government property does not include manufacturing defects, obsolescence, normal wear and tear, or purposeful destructive testing. Unless otherwise stated in the contract/order, loss of Government property does not include normal and reasonable inventory adjustments, i.e., losses of low priority consumable material, such as common hardware, as agreed to by the contractor and the Government Property Administrator.

3.5.2. Reporting Loss of Government Property

The contractor shall report loss of Government property in the GFP Module located in PIEE, <https://piee.eb.mil/piece-landing/>.

3.6. Return or Retention of Government Property

All property provided by the Government remains the property of the Government and shall be returned to the Government as directed, but no later than completion of the contract/order, unless the Procuring Contracting Officer (PCO) directs the contractor to retain the property for continued use under a successor contract. All Government property shall be returned to the Government in the condition provided unless approved in advance by the PCO. All material generated under the contract/order becomes the property of the Government and shall be returned to the Government as directed, but no later than completion of the contract/order, unless the PCO directs the contractor to retain the material for continued use under a successor contract.

4. TECHNICAL REQUIREMENTS

4.1. Period and Place of Performance

4.1.1. Contract/Order Period of Performance

The period of performance for the contract/order will extend 14 JAN 2026 through 13 JUL 2027.

4.1.2. Hours of Operation

Contractor personnel shall perform work during normal business hours Monday - Friday, 0700 - 1700 local time, excluding 11 Government-observed holidays, which currently include:

New Year's Day	Labor Day
Martin Luther King Day	Columbus Day
President's Day	Veteran's Day
Memorial Day	Thanksgiving Day
Independence Day	Christmas Day
Juneteenth Day	

In addition to the days designated as holidays, the Government may observe the following days as Government closure days:

- Any other day designated by Federal Statute
- Any other day designated by Executive Order
- Any other day designated by Presidential Proclamation
- Any other day designated as closure day with regard to Robins AFB facilities

On Government observed holidays and days designated by Federal Statute, Executive Order, Presidential Proclamation, or as closure days, when a Government staff representative is not present onsite, contractor personnel will not be permitted to work within the Robins AFB facilities, unless approved by the Government PM or COR. For regular Government observed holidays telework shall not be permitted. For ad-hoc designated holidays or other closures, the Government intends to allow telework but will need to approve on a case-by-case basis depending on the circumstances of the closure and workload.

In the event that an urgent situation arises that requires contractor personnel to rapidly respond and perform during periods of facility closures, access to facilities will be provided by the Government to allow appropriate contract staff members to respond and take necessary preventative, corrective, and/or recovery actions. Situations typically requiring rapid response during facility closure are those associated with critical system failures, with environmental issues carrying the potential to damage essential systems or equipment or to render equipment warranties and/or maintenance invalid, and/or those that could result in data loss. While Government technical program staff and CORs associated with the program will be responsible for providing the contractor with access to Robins AFB facilities, the Government contracting officer will be responsible for providing direction to the contractor regarding any temporary suspension of performance.

The contractor may be required to perform work under this contract in excess of eight (8) hours per day. Therefore, normal business hours or alternate business hours may be altered for a specified or indefinite period by the appropriate contract program management staff.

4.1.3. Accommodation

The contractor is required to submit all requests for accommodation for review and final approval by the Government. Accommodation requests that impact the contractor's ability to perform work onsite will be reviewed by the PMO and a decision (approval or denial) issued within 10 business days. Work

approved to be performed at an alternative location will be subject to remote work conditions outlined in 4.1.6 of this document. The contractor shall provide a written request with appropriate rationale and justification for the Government Program Manager/COR approval prior to any accommodation agreement.

4.1.4. Location

These services will be performed at Robins Air Force Base.

4.1.5. Travel

Contractor travel may be required. Contractor travel will be billed as a cost reimbursable item and is subject to the guidelines outlined in the Joint Travel Regulation (JTR). Reimbursement of travel expenses may be subject to submission of expense receipts prior to invoicing for reimbursement. The contractor shall be responsible for making their own travel and lodging arrangements as necessary to support the requirements outlined. All requests for travel shall be approved by the Government Program Manager or COR prior to expending funds. A detailed trip report will be provided to the Government Program Manager/COR not later than five (5) business days after completion of travel. Receipts must be presented with the invoice for reimbursement of actual travel costs.

(Exhibit A: Exhibit A: CDRL A006, Report, Record of Meeting/Minutes, DI-ADMN-81505).

4.1.6. Remote Work

All contractor staff are required to work on-site at Robins AFB daily (per section 4.1.5) and must be geographically located nearby. Remote work may be considered on a case-by-case basis by the Program Management Office (PMO) for positions with difficult-to-fill skillsets. The contractor is required to submit all remote work requests to the PMO for review. Each request must clearly identify the specialized skillsets considered difficult-to-recruit and outline the plan for fulfilling all contractual obligations while working remotely. The PMO will review each request and issue a decision (approval or denial) within 10 business days. Work approved to be performed at an alternative location will require use of Government Furnished Equipment (GFE) which will be provided and authorized for use by the Government. All GFE used for remote work will be configured per requirements outlined within Air Force Manual (AFMAN) 17-1301, Computer Security (COMPUSEC) and National Institute of Standards and Technology (NIST) Special Publication (SP) 800-46, Guide to Enterprise Telework and Remote Access Security. Operating costs associated with the contractor using an alternative worksite will not be assumed by the Government and will be the responsibility of the contractor. The contractor shall maintain availability during core hours (per section 4.1.1) via email and phone. All expenses for internet, telephone connectivity, and other equipment/facility services shall be borne by the contractor. The contractor shall provide a written request with appropriate rationale and justification for the Government Program Manager/COR approval prior to any remote work agreement.

Classified work may be authorized at an alternate worksite with the approval of Government Program Manager or the COR and the security office only with an approved and assigned device ONE SecureView (ADSV) machine. If approved, all users are required to follow all Security Guidance for Classified Portable Electronic Devices guidelines to include the 'Acceptance of Risk to Operate United States Air Force (USAF) SecureView (SV)' memorandum published by the Defense Information System Agency (DISA).

4.2. Contract Data Requirements List(s) (CDRL)

CDRL	Title	Authority	PWS Reference	Date of First Submission
A001	Technical Report-Study/Services	DI-MISC-80508B	5.3, 5.4, 5.6, 5.10.1.10.1.3, 5.10.1.4, 5.10.2.1, 5.10.3, 3.1.1, 3.1.2, 3.1.3	As required
A002	Status Report	DI-MGMT-80368A	5.3, 5.5, 5.10.1.4, 5.10.4	40 days after contract award
A003	Software Transition Plan (STrP)	DI-IPSC-81429A NOT 2	5.10.3, 5.10.5, 3.1.3	As required
A004	Computer Program End Item Documentation	DI-IPSC-80590B	5.10.3, 5.10.5, 3.1.3	As required
A005	Conference Agenda	DI-ADMN-81249C	5.10.6	As required
A006	Report, Record of Meeting/Minutes	DI-ADMN-81505 NOT 1	4.1.5, 5.10.6	As required
A007	Program Management Plan	DI-MGMT-81797 NOT 3	5.10.4	As required
A008	Environmental, Safety, and Occupational Health (ESOH) Plan	DI-SAFT-82313	6.4.2	40 days after contract award
A009	Accident/Incident Report	DI-SAFT-81563 NOT 2	6.4.1	As required (within 24 hours of incident)

4.2.1. Quality of CDRL Deliverable

There shall be no more than one rejection of any deliverable. There shall be no more than two total rejection(s) of deliverables per month. The Government will reject a deliverable if one or more technical errors or three or more minor errors are found within the deliverable. A technical error is defined as the format not being in accordance with the CDRL or the content not being accurate and complete in accordance with the CDRL, PWS, or contract. A minor error is exemplified by a typographical error, a grammatical error, etc. The rejected deliverable shall be corrected and resubmitted within 10 business days of notification of Government rejection.

4.2.2. Receipt of CDRL Deliverable

CDRL deliverables shall be submitted via email to the COR and PCO and submitted directly to the MFT member identified on the CDRL. There shall be no more than one (1) late submission(s) of deliverables per contract year. The contractor shall notify the Government if the delivery of any data/document will not meet the scheduled delivery date and negotiate a revised delivery date. The data/document shall be delivered by the revised delivery date acceptable to the Government.

The contractor must participate in unit security training, initial and annual) in accordance with 32 CFR Part 117, National Industrial Security Program Operating Manual (NISPOM), § 117.7(a). Contractors shall complete and maintain records of all required annual training as directed by the Government.

4.3. Antiterrorism (AT)

Contracts must comply with the AT provisions of the Defense Federal Acquisitions Regulation Supplement, DFARS: 252.204-7004 Antiterrorism Awareness Training for contractors.

5. TECHNICAL SUPPORT

5.1. Technology Support

The current PDSS support environment is described in Appendix B. The contractor shall operate, maintain, and facilitate expanded usage through implementation of new data sets, expansion of the user base, and incorporation of new solutions within the Robins AFB product data environments. While the current hardware infrastructure has been designed to support expanded use and increased storage requirements, the infrastructure components are subject to change to ensure continued viability of the infrastructure to support mission requirements. The contractor shall ensure the product data infrastructure and processes remain viable tools for development, management, storage, sustainment, publication, and distribution of DAF product and technical data. These activities include development of new, or enhancement, or improvement of existing system implementations, and establishment and management of processes to extract, transform, and load data to ensure requirements associated with DAF enterprise-level system implementations are satisfied. The contractor shall provide services necessary to research hardware and software solutions, to provide purchase recommendations to the Government Program Manager, and to implement hardware and software changes within the environment as required. The contractor shall also be responsible for configuration, testing, and implementation of new hardware and/or software applications as necessary and shall routinely be required to implement changes in the environment as necessary to ensure compliance with mandatory security directives/updates, and to ensure performance parameter requirements are achieved. The contractor shall provide personnel capable of working independently and who possess demonstrated working knowledge of the hardware and software components within the PDSS environments to support DAF processes. The contractor shall be responsible for ensuring all training and certifications prerequisite to gaining access to, and performing system support functions requiring elevated access privileges on systems attached to unclassified and classified DAF and stand-alone networks and the supported PDSS infrastructure are obtained and maintained by contractor personnel performing duties requiring such certifications.

5.2. System Support Services

Technical support services include system architect, system administration, and database management services necessary to manage and sustain hardware infrastructure and software components utilized in the capture, storage, maintenance, electronic distribution, transformation, and editorial processes associated with product data, and to ensure system and data contingency infrastructure and operations are implemented, viable, and documented. The following is a list of major software products currently utilized for product data management and supporting processes:

- Technical Data-Product Data Management (TD-PDM) - TD-PDM is a collection of subsystems with Commercial-Off-The-Shelf (COTS) and Government-Off-The-Shelf (GOTS) components based on Product Lifecycle Management (PLM) technology. These subsystems provide users worldwide with single-source access to multiple linked data sources which also support View-On-Demand (VOD) and Print-On-Demand (POD) processes. The system provides storage, management, version control, and configuration management capabilities for electronic product

data (TOs, engineering data, et al). TD-PDM is supports over 10,200 users from the .mil and .com network. These are primarily personnel throughout the DAF community, but the user base also includes users from other Department of Defense (DoD) Services and contractor community. TD-PDM serves as the repository for digital engineering drawings and related technical data. These capabilities support DoD logistics business functions, such as maintenance, repair, procurement, and re-engineering by providing worldwide, on-demand access to digital engineering drawings and associated data. See Appendix B for a list of tools and applications used within TD-PDM.

- Technical Order Authoring and Publishing (TOAP) - TOAP is a collection of COTS and GOTS components providing a complete TO content management, sustainment, and publishing environment. TOAP is the designated DAF organic TO authoring and publishing component which employs workflow-controlled process solutions for TO management and editorial workgroups. TOAP stores and manages document and content components of TOs within a dynamic database repository, tracking changes against baseline TO documents and allowing users to reuse information objects. TOAP enforces authoring and publishing compliance outlined within DAF policy. See Appendix B for a list of tools and applications used within TOAP.
- Secure Technical Order Repository (STOR) - STOR is a classified system and is the overarching system title assigned to the various COTS and GOTS products utilized to provide document content management, process and management workflows, and complex data publishing capabilities for classified TOs. The STOR system also functions as the DAF repository for archive and retention of electronic copies of classified DAF TO and Time Compliance Technical Order (TCTO) documents and associated increments (e.g., changes, supplements). As such, this system is utilized to support weapon system programs throughout the DAF. See Appendix B for a list of tools and applications used within STOR.

5.3. System Sustainment

The contractor shall manage and sustain the Robins AFB product data infrastructure. This includes the identification, recording, tracking, and resolution of all problems/issues, to include documentation and cause of system downtime (whether hours or unscheduled), and maintenance of a historical database to facilitate rapid identification of common system problems and trend analysis activities. The contractor shall ensure configuration control of COTS and GOTS products, and customized tools/scripts, is maintained, employing tracking and versioning techniques necessary to allow traceability of details associated with modifications/upgrades. The contractor shall document and employ disciplined testing and implementation processes to ensure modifications are implemented with minimal impact to the user community and to prevent degradation of performance and processes.

(Exhibit A: CDRL A001, Technical Report-Study/Services, DI-MISC-80508B)

The contractor shall perform actions necessary to facilitate compliance with DoD-mandated data center consolidation and cloud computing efforts. The contractor shall perform actions necessary to foster the adoption of safe, secure cloud computing options, to include the implementation of organizational cloud computing environments, steps to achieve data center consolidation objectives, and delivery of cloud services.

(Exhibit A: CDRL A001, Technical Report-Study/Services, DI-MISC-80508B)

The contractor shall provide for the operation and maintenance of the PDSS infrastructure, supporting development, conversion, storage, improvement, and use of product data to facilitate business and operations, and maintenance activities until capabilities and data reside in the AF-PLM solution or other enterprise level system(s), and the systems supporting those capabilities are no longer operational at Robins AFB. The contractor shall monitor and manage assigned systems to ensure a 98.5 percent monthly system availability rate (based on a 30-day month, 100 percent system uptime would equate to 720 hours) to all users. Scheduled downtime and downtime caused by external factors (such as power and network outages) shall be documented and tracked but will not be used to calculate system availability for the purposes of this performance objective. The contractor shall provide support to users from 0700 to 1700 local time.

(Exhibit A: CDRL A002, Status Report, DI-MGMT-80368A)

5.3.1. Help Desk Sustainment

Types of Help Desk services to be performed include, but are not limited to:

- Configuring/managing workstations, printers, scanners, and other peripheral devices.
- Tracking GFE hardware and software inventory, ensuring utilization of software licenses does not exceed the availability of valid software licenses.
- Coordinating with Government personnel when Help Desk problems occur.
- Documenting user requirements for reports and queries.
- Reviewing user issues/requests, determining system/software impact, and ensuring timely resolution of user problems
- Developing and sustaining documentation associated with PDSS implementations, programs, scripts, and other automation tools.
- Providing user training. Support training development of external efforts, not just applications managed within PDSS portfolio, including but not limited to areas within LZP-managed workload and AF-PLM CloudOne solution development.
- Establishing and managing user privileges and profiles.
- Providing support to system users, resolving user trouble calls, and satisfying requests for assistance or information.
- Management and operation of multi-tiered technical help desk environment to support systems and applications within the PDSS environment.
- Providing day-to-day user and help desk support for receipt, analysis, corrective action, and tracking of trouble and/or assistance calls received from customers on systems supported.

5.3.2. Operations Sustainment

Types of Operations sustainment services to be performed include:

- Installing and configuring operating system and application software packages and updates and managing system configurations.

- Defining hardware, software, and cloud computing requirements, researching potential solutions, and providing recommendations for purchases and/or upgrades to the Government Program Manager.
- Tracking requirements associated with, and providing cost estimates for, maintenance of systems hardware, software, and sustainment of system support services (such as cloud computing).
- Coordinating and/or executing acquisition of required maintenance agreements in a manner preventing lapse/gaps in maintenance coverage required for systems/processes.
- Tracking server hardware and software inventory, ensuring utilization of software licenses does not exceed the availability of valid software licenses.
- Reviewing all Air Force Network Operations (AFNETOPs) task orders (e.g., Time Compliance Network Orders (TCNOs), Information Assurance Vulnerability management (IAVMs), Maintenance Task Orders (MTOs), etc.), determining applicability to the Robins AFB PDSS, and implementing those found to be applicable.
- Ensuring Government-mandated security measures are installed and operational on all systems to include contingency environments (e.g., virus protection, firewall, authorization policies, account management, etc.).
- Coordinating with Government personnel when Operations problems occur.
- Managing system security.
- Performing system backups and restores.
- Monitoring and tuning the operating system.
- Performing system troubleshooting and point of failure isolation and determining solutions for system problems.
- Minimizing system downtime, notifying and obtaining approval from the appropriate Government representative in advance of scheduled downtime, and notifying the appropriate Government representative immediately upon discovery of other restrictions to processing or system availability.
- Creating, monitoring, and tuning databases utilized to support storage, product data transformation, configuration management, sustainment, and distribution.
- Supporting the PDSS applications, contingency environment services, and repositories.
- Identifying system problems to appropriate system points of contact and participating in actions necessary for problem resolution.
- Reviewing/updating/documenting actions associated with DAF enterprise-level systems, legacy system deconstruction, and data center consolidation/migration.
- Generating system and/or data reports.

5.3.3. Cybersecurity Sustainment

- Coordinating with Government personnel when cybersecurity problems occur.
- Verifying system security.

- Maintaining continuous monitoring capabilities including: updating system documentation and artifacts; maintaining comprehensive hardware and software asset inventories; analyzing vulnerability scan reports; managing Plans of Action and Milestones (POA&Ms); completing Security Technical Implementation Guide (STIG) checklists in accordance with applicable standards and requirements; conducting audit log reviews and analysis; and performing audits of security-related tasks and controls to ensure compliance with established security policies and procedures.
- Gathering and input of data and development of documentation necessary to ensure satisfaction of requirements driven by Configuration Control Board (CCB), portfolio management (via Information Technology Investment Portfolio (ITIPS)), certification/authorization and accreditation (via Enterprise Mission Assurance Support Services (eMASS) and/or other designated systems), and National Defense Authorization Act requirements and mandates.

5.3.4. Development sustainment

- Configuring, documenting, testing, and implementing new or updated software applications utilized to support technical data processes
- Conducting compatibility testing to ensure continued operation of required product data applications on DAF standard desktop configuration releases.
- Performing software application testing and completing required documentation for submission of required PDSS applications to the DAF Evaluated Products List (EPL) program office for subsequent evaluation/approval.
- Coordinating with Government personnel when development problems occur.
- Developing and sustaining documentation associated with PDSS implementations, programs, scripts, and other automation tools.
- Performing system integration support.
- Web design and delivery of electronic data.
- Developing processes supporting product data transformation, management, sustainment, and distribution.
- Developing and sustaining of tracking applications and other electronic inventory and auditing systems.
- Assisting the Government in data input/integrity, quality assurance, and data migration processes.
- Participating in testing, scanning and implementation of new software releases.
- Developing and executing plans and actions necessary to facilitate DAF enterprise-level system (e.g., ETIMS, TOAP, and solutions resulting from the DAF Logistics Capability Initiatives) implementation/data migration.
- Providing support services necessary for sustainment of hardware and software utilized in product data capture, conversion, and clean-up, and scanning operations.
- Providing support services for hardware and software utilized in product data quality assurance operations.

- Providing technical system and database support services necessary for successful system operation.
- Providing the skills necessary to establish and maintain web-based access to, and for execution of queries and retrieval of product data from within supported systems and integrated third party commercial repositories. Supporting system implementations in both unclassified and classified environments.

5.4. Software Licensing/Testing

The contractor shall maintain and track all Government-owned software license documentation and media for non-standard software utilized within the PDSS environments, shall ensure that all software installed is utilized in accordance with appropriate license agreements, and that software managed is processed for approval and placement on the DAF EPL or within the appropriate certification/authorization and accreditation package. The contractor shall maintain and utilize a software test and evaluation enclave compliant with DAF requirements for performance of software testing. The contractor shall conduct client/server testing in accordance with the most current HQ CCC/Software and Application Certification Assessments (SACA) (or other governing authority) software certification processes and governing Air Force Instructions (e.g., AFI 17-101). The contractor shall utilize automated tools to perform software usage audits and shall provide quarterly reports to the Government outlining software utilization metrics, and status of EPL approval for COTS and GOTS software not reflected, or approaching expiration on the DAF EPL.

(Exhibit A: CDRL A001, Technical Report-Study/Services, DI-MISC-80508B)

The contractor shall be responsible for identifying, resolving, and reporting instances of usage of illegal software on assigned systems. Instances of illegal software installation shall be resolved or a waiver submitted for Government approval within 30 days of identification. The contractor shall promptly identify requirements for additional software and/or licenses to the Government Program Manager as required for new capabilities, to resolve licensing issues, and/or to ensure continued support of products under projected configurations of DAF standard desktop and server configurations.

(Exhibit A: CDRL A001, Technical Report-Study/Services, DI-MISC-80508B)

5.5. 'Warm' Site and System Backup Capabilities

The contractor shall be responsible for backup and restoration, as required, of both system and data files for all systems within the PDSS environment. The contractor shall develop and execute all backup-related processes necessary to ensure restoration of systems and data can be accomplished as required. Backup processes shall include the execution of both incremental and full back up routines, with incremental routines executed daily and full back up routines executed weekly, at a minimum. Backup routines shall produce multiple copies of backup media, to allow one (1) copy of backup media to be stored on-site within provided fire-proof containers in the Building 301 classified vault and one (1) copy to be stored within a General Services Administration (GSA) approved container in a Government Program Manager designated building on Robins AFB. The Government Program Manager/COR will certify on a monthly basis 100 percent synchronization and availability of data transferred to the continuity of operations (COOP).

The contractor shall execute processes necessary for storage and retrieval of both unclassified and classified (up to and including SECRET) PDSS backup media according to the Government-approved

rotation plan. The contractor must provide personnel appropriately authorized by the Government security manager to courier classified data in order to transport all classified backup media to/from the storage locations.

The hardware and software comprising the current COOP infrastructure is depicted in Appendix B. The contractor shall perform efforts to manage and sustain infrastructure and processes associated with support of the hardware and software required for backup of critical product data at the contingency location. These efforts shall include all actions necessary to ensure continued viability and support of hardware, software, and network connectivity; and those actions necessary for development, sustainment, and execution of plans and processes to perform data replication and periodic data synchronization processes. The contractor shall ensure critical data from the production systems at Robins AFB are replicated within the 'warm' site backup systems location daily such that all data for critical systems is available at the 'warm' site location. The contractor shall certify successful replication of data (within 'warm' site systems and/or on media located at the 'warm' site location) on a monthly basis.

(Exhibit A: CDRL A002, Status Report, DI-MGMT-80368A)

Routine administration of the system at the 'warm' site should typically be accomplished by contract personnel located at Robins AFB utilizing remote administration methods with synchronization of data routinely accomplished via an automated update process. Contractor personnel will be required to perform system support activities in the event of hardware failure or if network connectivity is unavailable and shall be responsible for backup of the 'warm' site equipment. In the event a catastrophe occurs, which results in destruction of the primary systems at Robins, contingency plans to utilize the system at the 'warm' site must be in place to allow the 'warm' site systems to be utilized for access to critical data. The contractor shall identify to the Government Program Manager the key personnel required to deploy limited system and user support activities for critical technical data systems at the 'warm' site location. Key personnel will be considered mission essential (reference paragraph 6.10) and shall be required to travel to and perform system and user support efforts at the 'warm' site location, if able, after catastrophic events.

5.6. System User Training

The contractor shall provide user training for Government and contractor employees as required by the Government Program Manager. The training shall cover general system functionality and operation, in-depth functional training based on specific user roles/responsibilities, and data maintenance/improvement operations training for those individuals responsible for data configuration, cleansing, and accuracy both internal and external to the PDSS portfolio. 'Train the Trainer' type training will be utilized to the extent possible to support a select group of Government and/or contract employees who may provide system support and/or initial and recurring training to system users.

(Exhibit A: CDRL A001, Technical Report-Study/Services, DI-MISC-80508B)

5.7. Production Publishing Support

The contractor shall support the production publishing tools and support processes implemented within the TOAP environment and shall facilitate automation of additional business and sustainment processes within those tools. This includes customization and configuration of systems to support the unique functions of various programs, and to insert and distribute technology improvements. Prior to

development or insertion of new or improved processes or capabilities, the contractor shall gain approval from the Government Program Manager and shall ensure developed capabilities are fully tested and documented (to include installation and operation procedures), and that impacts to the system's Authority to Operate (ATO) are assessed and addressed via the appropriate processes.

The contractor shall also provide support for the publication toolsets and content management system by resolving issues identified by customers via help desk ticket submission, by executing content management system schema and workflow planning and design, and by scheduling and providing toolset training for customers, to include training on utilization of Government-owned and maintained FrameMaker templates and on proper methods of tagging technical data in Standard Generalized Markup Language (SGML) in order to reduce issues upon data delivery. The contractor shall support product data electronic repository, and publishing systems comprised of relatively complex COTS content, configuration management, and publishing software applications implemented to support military-unique processes and publishing requirements. Contractor shall be knowledgeable of documents governing the configuration control, storage, distribution, creation, and sustainment of military product/technical data and the software tools and processes utilized to compose technical publications from structured/tagged source data. The contractor shall possess the skills necessary to develop and sustain the tools necessary to transform structured/tagged source data and associated graphics files into military- and international- specification compliant output formats utilizing XPP, Live Content, or other COTS publishing environments. The contractor shall evaluate changes to directives governing product data content structure in order to assess impacts to the production publishing toolsets and to the data sustained within the content management and associated data within other technical data management systems.

5.8. Content Management System Implementation/Support

The contractor shall assist in efforts to expand the implementation of COTS and GOTS technology designed to provide product data content management, sustainment process management, and configuration control capabilities. The contractor shall populate and aid in the sustainment of management data, shall develop and sustain required workflows, provide user and system/database administration training (as required), and shall add and manage system user accounts. The contractor shall be responsible for all system management, sustainment, and configuration management aspects of the content management subsystems. The contractor shall assist the Government in efforts to analyze various content management and associated publishing system interfaces and capabilities and shall perform efforts necessary to assist the Government in consolidation of those systems, capabilities, and data.

The contractor shall work with the Government technical program staff to determine specific content management system implementation/expansion activities, data migration and workflow requirements, and shall be responsible for gathering and documenting the details and schedules associated with those requirements. Efforts to populate the system with the necessary management data shall include migration of data from legacy sources. Electronic transfer/mapping and automated migration of data shall be utilized to the fullest extent possible. However, some manual entry of data into the system/subsystems may be required.

The contractor will be provided access to obtain the following types of data from Government data sources to aid in population of the content management system:

- A listing of type and quantity of technical data managed, to include other pertinent information about the data such as change or revision level, change or revision date, and managing organization.
- A listing of technical content management and sustainment personnel (if applicable) associated with specific types of technical data.
- Access to Government source data files.

The contractor shall develop and implement a method to ensure data migrated into the system accurately depicts the most current available configuration of the source data utilized to populate the system. This may include comparison of data within multiple data sources. The contractor shall work with Government technical and functional Subject Matter Experts (SME) to determine the best sources of data for use within the system.

5.9. S1000D Specification Data Capabilities

The DAF has mandated that programs acquire Type 2 Interactive Electronic Technical Manuals (IETMs) authored to the international technical data specification S1000D utilizing DAF business rules. Several weapon system programs are currently pursuing transformation of data from legacy formats (SGML, Portable Document Format (PDF), et al) to the S1000D specification structure. The contractor shall perform actions necessary to ensure existing COTS solutions for management and sustainment of S1000D data remain viable and shall implement and maintain capabilities necessary to facilitate development, receipt, validation, storage, management, sustainment, rendering, and distribution of technical data content authored to the S1000D specification.

The contractor shall review the DAF S1000D business rules and identify deficiencies, limitations, and/or impediments to mission objectives inherent within the business rule document set. The contractor shall work with Government technical program staff and other contractors involved in efforts to mature the DAF business rules and to develop project-level business rules and shall develop documentation necessary for the Government Program Manager to initiate requests for business rule modifications as required to achieve program objectives.

The contractor shall perform efforts to integrate/implement and/or sustain required S1000D data capabilities (utilizing COTS tools) within the existing Robins AFB content management and publishing environment. The contractor shall work with the Government technical program staff and other customers to develop and document processes and workflows to be utilized by authors, editors, and program/publication managers for the creation, management, sustainment, and delivery of technical information structured in accordance with the S1000D specification. All capabilities and processes developed/implemented shall include measures designed to ensure data introduced and sustained within the S1000D solution set is in compliance with the S1000D specification and the business rules adopted by the DAF and/or specific DAF program.

Review of the DAF business rules and implementation of S1000D support capabilities should be conducted such that, to the greatest extent possible, rendering of multiple output formats can be achieved from a single set of source data. For example, the contractor should evaluate the potential of utilizing existing content management and publishing tools to facilitate automated transformation of existing structured TO content data (SGML) into data type supported by the S1000D specification (XML) and mapping transformed content into S1000D data modules to allow rendering of both page-based and Type 2 output from the same source data.

The contractor shall assist the Government in assessing COTS and GOTS S1000D viewing solutions. The contractor shall provide services necessary to research available products, determine level of product cybersecurity compliance (as is necessary for operation within DAF environments), assess ability of product to integrate with external systems, and assess functionality of product solution against documented DAF functionality requirements. The contractor shall produce a report detailing the methods utilized for assessment and depicting results of each product assessed.

5.10. Subject Matter Expert Support

The contractor shall provide personnel who possess the functional and technical expertise necessary to support execution of the Robins AFB PDSS Program. Subject matter expertise shall be provided by the contractor as necessary to perform management activities associated with sustainment of the technical environments, data migration, improvement processes, and information software security support as outlined within this PWS. The scope of services to be provided by subject matter expert personnel includes, but is not limited to:

- Gathering and documenting user and technical requirements and translating those requirements into system specifications, data management plans, configuration management plans, test plans/scenarios, life cycle management, and other related documentation.
- Participation in functional and technical meetings, conferences, and reviews related to system and data process requirements.
- Gathering information from users, identifying work problems, and designing system or processes to resolve problems or satisfy data requirements.
- Providing information assurance expertise as it pertains to supported systems, to include support of contingency planning, system security testing, and annual contingency exercises.
- Providing input into functional, process, and data modeling activities in support of planning and analysis efforts.
- Coordinate/facilitate annual Tabletop or Live Contingency/Incident Response Plan. Examining and analyzing current and contemplated operations as necessary to develop and implement system, process, and/or data improvements.
- Utilizing data design, proposed system revisions, and test results to ensure functional requirements are met.
- Researching, analyzing, and providing system, functional, security, and data information required to assist the Government technical program staff in satisfying requests for information.
- Provide expertise related to Federal Information Security Management Act (FISMA) compliance requirements.
- Provide expertise necessary for analysis and development of Plan of Action and Milestone (POA&M) documents as required.
- Provide expertise necessary for analysis of military and commercial specifications governing product data, and for assessment of impacts of specification changes to supported systems and data.

- Risk Management Framework (RMF) expertise for creating and maintaining comprehensive RMF packages to achieve and sustain Authority-To-Operate (ATO) certification for STOR, TD-PDM, and TOAP systems, including authoring, reviewing, and maintaining all required documentation and system artifacts such as: Standard Operating Procedures (SOPs); Ports, Protocols, and Services Management (PPSM) documentation; System Security Plans (SSPs); Security Test Plans (STPs); hardware and software inventories; Information Technology Categorization and Selection Checklists (ITCSCs); and all additional documentation required for RMF compliance and ATO approval in accordance with applicable federal standards and organizational requirements.
- Conduct testing of security controls and provide detailed results, mitigating factors, test scripts, and other required evidence for each applicable sub-control in eMASS, as directed by the Authorizing Official (AO), Security Control Assessor/Security Control Assessment Report (SCA/SCAR), Program Manager (PM), or Information Systems Security Manager (ISSM).

Contractor personnel satisfying subject matter expert requirements shall be individuals with a specific level of expertise in certain, key system or product data areas or disciplines, who are also knowledgeable of the domain of supported systems. These professionals shall provide unique and/or in-depth functional and/or technical knowledge associated with product data, supporting applications, system operations, and cyber security. Contractor personnel satisfying a subject matter expert role should not require specific application training funded by the program or DAF on PDSS applications in order to support workload. Contractor shall identify alternate SMEs for each system/project/workload area to maintain system/project/solution momentum. Alternate experts will be designated to provide coverage during primary SME unavailability eliminating any single point of failure. The contractor shall provide a list of these alternate SMEs to the COR or Government PM.

5.10.1. Transformation and Migration Support Services

5.10.1.1. Digital Data Strategies

The contractor shall provide services to support the development and execution of digital data support strategies, system capability, and data extraction, transformation, and migration/loading efforts. The contractor shall perform analysis and reporting activities and provide technical assistance necessary to support the Government in conveying requirements to facilitate transition of system capabilities and data to DAF enterprise-level product/technical data solutions.

(Exhibit A: CDRL A001, Technical Report-Study/Services, DI-MISC-80508B)

Supporting digital data strategies requires staff possessing expertise in data standards, conversion, migration, transformation, and management activities, and expertise in the PDSS capabilities employed to enable and manage those activities.

The contractor shall support efforts associated with analysis of TO data managed by Robins AFB, et al., organizations in order to categorize the current state of the TO source and output data formats, and shall perform efforts to develop/mature a decision tree and business case analysis toolset to aid data management organizations in making data conversion decisions. Data analysis requirements include but are not limited to:

- Determining data type analysis requirements to include categorizations and classes of data.

- Developing a profiling methodology including tools and technologies to be used in data gathering.
- Developing an approach for identifying all TO programs and assuring inclusion of all TO data.
- Performing data gathering including site visits and surveys.
- Where structured data (i.e., SGML or XML) is available, utilizing automated scripts that mine data instances to determine specification compliance, tag usage and ways in which elements are being used.
- Surveying content creators to gain an understanding of source data authoring and management considerations, e.g. which source formats and governing Document Type Definitions (DTDs) or specifications (including versions) are used, the volume of data written to these DTDs or specifications, and problems encountered in developing the data using the structures governing source data formats.
- Developing a data profiling and reporting methodology.
- Developing reports and recommendations based on analysis results, including optimization paths for each data type.

5.10.1.2. Technical Capabilities and Data Transformation/Migration

The contractor shall develop or enhance existing publication system style bundles and templates necessary to produce military-specification compliant page and/or web-based TOs from structured and unstructured source data formats, and to produce non-page-oriented output of S1000D data sets for presentation to customers in IETM form. Contractor activities shall also include efforts to identify and support the use of new and emerging technologies that will enhance the data publication and management processes and allow migration of standard data in legacy formats to newly adopted DAF data standards.

5.10.1.3. Product/Technical Data Quality Assurance and Data Loading

The contractor must ensure documents delivered from editorial, conversion, development, enhancement, and sustainment processes undergo a quality review to ensure compliance with applicable specifications and/or outlined quality standards.

The contractor shall perform quality assurance reviews by random sample on PDF files delivered for loading into the repository. Sampling processes shall be performed daily and shall ensure that five (5) percent of data delivered for loading is reviewed and that review results are documented and made available to Government technical program staff and data owners as requested.

Quality review processes shall include execution of processes to review TOs to ensure PDF files are adequate to support both print- and view-on-demand processes. Quality assurance processes associated with PDF files shall include, but are not limited to, execution of the following:

- ‘A’ page review of document submissions to ensure completeness.
- Verification of superseded notice, distribution statement, and export control notice, and destruction notice on title page of document.

- Verification of accuracy of list of effective pages, table of contents, list of tables, list of illustrations, and appendices.

PDF data quality assurance processes shall ensure TO text and graphic data is viewable and printable, pages are free of punch or drill hole shadows, contain no obstructions to content, are not rotated, and that contrast between background page color and foreground text/graphics is adequate to support print-and view-on-demand processes. Documents selected for quality review shall be inspected at 100 percent for format, view and printability, but not for verification of content, and shall be collected such that batch loading of files into the Product Data Management (PDM) Gateway may be accomplished.

The inspection process shall be documented in such a way that it may be reviewed and verified by the Government technical program staff. Files not meeting the inspection requirements shall be returned to the submitter with an explanation of reason for rejection. The contractor shall track common patterns of the rejections and recommend process improvement opportunities to the Government. Final acceptance of all captured images shall rest with the Government.

(Exhibit A: CDRL A001, Technical Report-Study/Services, DI-MISC-80508B)

The contractor shall perform activities necessary to load (typically in a batch load process) all received/inspected documents into the current vaulting system (or an approved alternate vaulting system) for access, query, viewing, storage, printing, and distribution of the electronic TOs. The index-attributes (listed below) shall be captured for each electronic document to be stored in PDM Gateway within the Teamcenter PLM application. The associated PDM Gateway database shall also contain a reference to each electronic TO file and its permanent storage location/file name.

Technical manuals may be collated to any change level in ascending change level sequence. The contractor shall capture data to populate the data elements outlined below for each Interactive PDF (I/PDF) file to enable users to access the documents within the electronic repository by a variety of search attributes (TO number, title, date, etc.).

Attributes of each TO shall be captured such that, after data load, metadata within PDM Gateway contains the information outlined below:

Attribute List		
Document Type	Formerly TO #	Bleed Borders
TO Identification	TCTO Data Code	Color
TO Pub Date	Title	Halftones
Change Number	Abbreviated Title	Paper Color
Change Date	System Type	Page Size
Safety Supp #	Series/Model #(s)	Page Count
Safety Supp Date	Part Number(s)	Longest Foldout
Operational Supp #	Reason	Special Instructions
Operational Supp Date	Category	Proponent Org
TP/TOPS #	Distribution Code	Routine Supp #
TP/TOPS Date	Approved	

An update process shall be used by the contractor to incorporate new changes and supplements released after a document has been initially loaded into the repository. This shall include the incorporation of TO updates into the vaulted TO, such that the vaulted document is the most current configuration of the

document. The contractor shall utilize existing or alternate processes approved by the Government Program Manager to ensure the most current version of documents stored in the vaulting system are available for user access and shall ensure previous versions are maintained in 'archive' status until no longer required.

The contractor shall utilize existing or an alternate approved process to facilitate expedited quality assurance review of TO documents which are needed to meet immediate print and or distribution requirements.

Delivery of processed documents shall be considered complete when the contractor has executed processes to load delivered/reviewed data into the appropriate electronic repository. The contractor shall complete the loading of PDF documents within seven (7) working days of receipt for documents submitted as routine priority, and within three (3) working day of receipt for documents submitted as urgent or emergency priority.

Quality review processes to ensure that structured editorial source data parses against applicable document type definitions upon import to the content management system shall also be performed by the contractor. Failure of the document to pass will require the contractor to analyze the document to determine point(s) of failure within the structured document. The results of the quality review will be routed to Government technical program staff, data managers/owners, and individual responsible for editorial action on the document.

The contractor shall support Robins AFB and other customers in performance of actions necessary to review and test structured source data received from other vendors to ensure compliance with appropriate standards and to ensure that document parses and output can be rendered from assigned publishing systems. If submission fails to pass compliance review and/or parse and render output, information regarding cause of failure will be documented and provided to the customer to facilitate correction by the vendor submitting the data files. This includes a Centralized Technical Order Repository (CTOR) Monthly Load Activity Report. Files not meeting the inspection requirements shall be returned to the submitter with an explanation of reason for rejection. Final acceptance of data load quality shall rest with the Government.

(Exhibit A: CDRL A001, Technical Report-Study/Services, DI-MISC-80508B)

5.10.1.4. Data Migration/Improvement Processes

The contractor shall provide services for the Parts Configuration Management System (PCMS) capture and sustainment effort to improve parts configuration data extracted from Illustrated Parts Breakdown (IPB) TOs resident with the PCMS and associated data load files. This includes services necessary to execute processes to convert, validate, identify errors, and perform initial data load to ensure data extracted from any newly identified IPB TOs is resident within PCMS, and within the other similar databases (i.e. Applications, Programs, and Indentures (API/D200F)) when required by the data management organization. The contractor shall utilize data enhancement toolsets to provide services to sustain and enhance parts configuration data already resident within PCMS database and/or associated data load/configuration files to ensure data accuracy and currency, and to ensure data structure and element requirements are satisfied. The contractor shall recommend and incorporate modifications to the current PCMS sustainment/capture/conversion processes and, after approval from the Government Program Manager, shall employ these changes to enhance data and/or data elements captured.

(Exhibit A: CDRL A001, Technical Report-Study/Services, DI-MISC-80508B)

Process improvement opportunities to improve sustainment/capture/conversion processes include, but are not limited to, using the TOAP SGML and XML output files to generate a baseline captured IPB list and then converted into the appropriate format for data formatting for importing into PCMS.

Implementation of approved modifications shall include actions necessary to ensure that any discovered or derived data elements not originally published in the source IPB are stored as XML tags in the source files and database and are viewable via the appropriate user interface(s).

The average number of parts per IPB is approximately 1,000 (when all captured IPBs are considered). The current sustainment activity scope is estimated to yield up to an average of 45,000 parts to be processed (loaded and available for use) each month. The contractor shall conduct an analysis of IPBs/updates to be processed into PCMS/D200F, shall prepare necessary work plans, and provide access to work plans/products to Government technical program staff and Contracting Officer Representative (COR) personnel. The contractor shall maintain and refine the work plan as needed to ensure successful processing of parts configuration data and load of PCMS/D200F applications.

(Exhibit A: CDRL A002, Status Report, DI-MGMT-80368A)

It is expected that data necessary for execution of PCMS data and process sustainment, enhancement, and execution support shall be available for contractor performance immediately upon award of order. All data shall be supplied to the contractor in electronic formats with the majority of the data in PDF-formatted documents. Data may also be supplied in database format for those systems that utilize IETM systems.

The contractor shall:

- Download IPB TOs and/or updates from CTOR or other electronic repositories/data shares for processing.
- Capture parts configuration data, to include Usable On Code (UOC) translation tables, from IPBs via sustainment, capture, and/or extraction processes.
- Process data improvements/additions utilizing established business rules tools/processes, utilizing an approved data set to facilitate data comparisons and to ensure currency and accuracy.
- Add additional data elements as required.
- Create the maintenance parts list (MPL) file for each IPB processed in the formats required by data load utilities.
- Utilize data load utilities to upload these MPL files into the PCMS application or other bill of materials associated applications as directed by the Government technical program staff.
- Refine and maintain the process documentation as needed, to include training materials.
- Create D200F loadable files and provide those files to D200F support personnel when required by policy or as requested by the technical content manager.
- Merge recap reports with the PCMS generated load files and associated part errors with the appropriate TO Figure and Index.

- Identify and resolve errors during bulk loading.
- Perform quality assurance functions to verify load file accuracy.

The contractor shall capture the contents of the MPL section of the IPB using previously developed/enhanced processes. If required DAF enterprise-level system attributes are missing within the source documents, those data elements will be returned in process records as appropriate null values. The contractor shall adopt American National Standard Institute (ANSI) Statistical Quality Control standards, to include the most current version of ANSI Z1.4.2003 to ensure that required data accuracy rates of 98 percent or greater per MPL are being achieved. The contractor shall provide a Parts Configuration Management System (PCMS) Monthly Load Accuracy Report. Files not meeting the inspection requirements shall be returned to the submitter with an explanation of reason for rejection. Final acceptance of data load quality shall rest with the Government.

(Exhibit A: CDRL A001, Technical Report-Study/Services, DI-MISC-80508B)

The contractor shall utilize existing (or, upon approval, enhanced) automated rules-based processes to compare and verify the accuracy and currency of captured data and data extracted from other electronic parts indenture and supporting databases. The contractor shall thoroughly test data throughout the indenture cleansing process for anomalies to ensure accuracy of output products and information.

The contractor shall produce a MPL file for each IPB or update and process for loading into the PCMS/D200F applications. Parts loaded into D200F shall be tracked and error reports produced to facilitate resolution of inaccuracies of parts not successfully loaded. On a monthly basis, the contractor shall execute PCMS semi-automated batch load process, to transmit a D200F/API load file containing records/updates associated with IPBs identified for D200F loading by the appropriate technical content manager. The D200F load process will generate a log file/recap report for records contained in the load file and shall outline load errors. The error rate will vary depending on the quantity and quality of the parts indenture data in the source IPB TOs. The contractor shall analyze the D200F load report to identify problem TOs and shall recommend appropriate corrective action.

(Exhibit A: CDRL A001, Technical Report-Study/Services, DI-MISC-80508B)

The contractor shall include PCMS program status within the monthly status report. Status report shall show progress, current status of associated projects, tool sets, and products, and shall quantify source and processed data, as well as errors and anomalies encountered during the sustainment and parts data loading processes and shall provide recommended approaches for issue correction.

(Exhibit A: CDRL A002, Status Report, DI-MGMT-80368A)

The contractor shall manage the processes necessary to initially populate and to update the PCMS/D200F applications with contents of the generated MPL files. The processes shall include loading and linking of data within the PCMS applications, and creation/submittal of D200F loadable files. Part indenture data that has not been verified will be flagged with appropriate error codes as unresolved, until such time as Government technical program staff or other contract personnel resolve outstanding discrepancies and/or errors.

The contractor shall sustain/enhance and execute processes to establish and sustain data relationships associated with PCMS data as necessary to link commodity TOs to respective aircraft TOs.

The contractor shall also deliver transformed PCMS data using the appropriate data cleanup tool sets and processes needed to:

- Incorporate the attributes required to support DAF enterprise-level systems.
- Establish indentured relationship between all related part and next higher assembly (NHA) data.
- Associate/link NHA/parts data in commodity TOs back to end item in associated aircraft TO.
- Create indentured Bill of Materials (BOM) structure that represents a complete and accurate representation of IPB source data utilized in data processing.

The contractor shall track part records processed and shall make tracking system, output products, and all associated documentation available for access by assigned Government technical program staff and COR personnel.

5.10.1.4.1 Engineering Drawing Relationship (EDR) Processing

Associations between the part indenture data within PCMS, referred to as BOM data, and the engineering drawing and computer-aided design and manufacturing data relevant to the part data in the BOM have been established. This data association enables the processes necessary for system configuration baselines to reside within a PLM system. These configuration baselines are to be sustained throughout the system lifecycle via the configuration management process implemented within a PLM system. In order to prepare data for migration to, and sustainment within, enterprise-level systems, the contractor shall perform efforts to develop, document, and execute processes to:

- Record and sustain the relationship between the captured PCMS BOM data and the All-Purpose Tool for Engineering, Analytics, and Management (A-TEAM) engineering drawings and CAD/CAM source files that are relevant to the parts data, and document any anomalies discovered during data processing. It is expected that the following steps will be required to accomplish the task objectives:
 - Extract the configuration data (commonly called the MPL) from the PCMS application.
 - Utilize PCMS and A-TEAM data to discover and analyze engineering drawings which may be associated with parts in the end item part indenture.
 - Determine and capture association points.
 - Capture association within the MPL files.
 - Load the enhanced configuration data (part indenture and associated drawing data relationships) into the directed PLM system.
 - If new drawing relationship tags cannot be identified based on the A-TEAM drawing loading solution, then the existing “YES” tags already determined should be used to sustain the tagged information in PCMS.
 - New revisions of drawings do not need to be loaded into PCMS and associated to the tags.
 - Drawings do not need to be made available in the PDM Gateway

The processes executed by the contractor must include a sequence of discrete tasks; documented and enabled by computer-aided functionality; must be repeatable, scalable, and production-oriented; and

must support iterative enhancement of the configuration. (Processes must initially support association of data at a high level of indenture and support refinement of the data associations to lower levels of indenture over time.)

Based on priorities established by the Government Program Manager, the contractor shall assist the Government technical program staff in defining data association requirements, and shall develop, document, and execute processes necessary to establish and maintain association (linking) of various types of product data within a PLM environment.

5.10.1.4.2 Full Text Content Search (FTCS)

The FTCS data conversion processes utilize a combination of COTS and GOTS software products within the A-TEAM application to perform automated extraction and transformation of multiple related raster images into a single output format with properties enabling enterprise-level search of the textual content of the raster image. During this conversion process, the textual content (dimensions, annotations, parts lists, call outs, and other character data) on the drawing image are made searchable and are captured within an optimized search index. The process and resultant files and index data are commonly referred to as FTCS.

Additionally, all TOs within PDM Gateway are processed to enable search of the textual content within the TO. The contractor shall execute the processes necessary to update and/or regenerate (as required) a keyword index. Enterprise search engine from the source content consisting of both the FTCS PDF documents and the PDM Gateway PDF documents (under configuration control within one (1) or more PLM environment vaults).

The contractor shall execute processes necessary to enable and sustain FTCS of source data and shall enhance tools and processes as required to ensure continued viability of the existing FTCS capability. The contractor shall fully document, test, and validate currency of the solution design outlining the end-to-end process and any changes as they are implemented. The contractor shall be responsible for execution of all processes necessary to convert and upload files and make searchable in the appropriate data repository. The contractor shall apply the necessary sustainment changes to the processes necessary to maintain the integration of the keyword index search with the existing applications.

5.10.2. Enterprise Level Capabilities/Data Support

5.10.2.1. Enterprise-Level System Support

Many DAF legacy system capabilities and associated data are to transition to the enterprise-level systems such as the solution expected to result from AF-PLM and from efforts to consolidate the DAF TO content management, authoring, and publishing system solutions. Over time, much of the functionality currently resident in Robins AFB PDSS is slated to incrementally transition to enterprise systems such as that resulting from AF-PLM and from other data center and system consolidation efforts. The contractor shall support the Government in developing, reviewing, and updating documentation associated with efforts to replicate, deconstruct, and/or enhance existing PDSS capabilities and processes within DAF enterprise-level or consolidated systems. The contractor shall also perform actions necessary to transform product/technical data and/or to map data to formats and structures necessary for successful transition into DAF enterprise-level and/or consolidated systems. The contractor shall provide expertise on product/technical data systems capabilities, data standards, and processes necessary to support replication/utilization of capabilities within planned or newly implemented DAF enterprise system environment(s).

(Exhibit A: CDRL A001, Technical Report-Study/Services, DI-MISC-80508B)

The contractor shall provide expertise required to restructure electronic TO data to meet the document structure requirements necessary to replicate Robins AFB electronic TO data in ETIMS. This includes development and execution of processes necessary to combine individual components of a publication into a single source file and link data within the component pieces, ensuring compliance with requirements outlined in applicable DAF governance documents. The contractor shall also manage and execute processes necessary to route files for approval by the cognizant TO manager and upload into the ETIMS electronic TO repository on-ramp upon receipt of approval.

The DAF TOAP enterprise solution was a major initiative designed to reduce the number of organic content management systems from five to a single system. Management and sustainment responsibility will transition as directed by AFMC/A4F/A4N upon completion of a migration to CloudOne, while infrastructure support will be maintained within the PDSS environment. Therefore, the level of technical systems and data support requirements are expected to be sustained and then potentially increase if/when DAF programs onboard to TOAP.

The contractor shall provide technical services to assist the Government in the management of DAF enterprise TOAP solution and associated data. Specifically, the contractor shall provide the technical support services necessary to plan and execute actions to extract, transform and load data, to add program-specific schemas, to add user accounts, and to provide user training to users transitioning from other targeted systems, to modify the TOAP solution tools to support additional military specification requirements driven by data migrating into the TOAP environment, and to assist customers in efforts to migrate data from other target systems into the TOAP environment.

The Integrated Data Information Manager (IDIM) is an LZP managed implementation within the Teamcenter PLM environment. IDIM provides integrated workflow processes and configuration management capabilities for linking engineering drawings to contracts, Contract Data Requirements Lists (CDRLs), weapon systems, data requests, and data deliveries. The system capabilities provided currently support management of tasks and documents required by DAF and DoD technical data acquisition and management policy/guidance. The contractor shall maintain the existing IDIM environment capabilities until the solution can be migrated into the AF-PLM CloudOne instance and activities performed in IDIM are no longer required.

These capabilities are utilized to manage engineering data support processes which are common to all three AFLCMC/LZP locations.

The contractor shall be responsible for management and sustainment of the system environments, to include those components added to the environment through consolidation of the targeted systems.

The contractor shall be responsible for management, operation, sustainment, and enhancement of the IDIM application, to include transition of the application necessary to ensure implementation of capabilities within approved COTS product versions, utilizing processes defined for support and sustainment of all LZPPS-managed product data systems.

The contractor shall provide technical services to assist the Government in the management and sustainment of the enterprise-level PLM system implementation within A-TEAM. AFLCMC/LZP is charged with responsibility for the technical consolidation effort, and retained program management, system sustainment, and application sustainment responsibilities for the newly expanded A-TEAM

implementation within TD-PDM, with functional ownership for that system implementation assigned to AFLCMC/XP-OZ. The contractor shall provide the technical support services necessary to plan and execute actions to support DAF PLM Capability Support Office (CSO) integration, known solution enhancements, and version migrations as required. The contractor shall provide process improvement recommendations, data clean-up support, and solution design enhancements to the A-TEAM environment until it can be migrated into the AF-PLM CloudOne environment.

5.10.2.2. Data Research and Mapping Support

The contractor shall support requests for information by outlining, developing, and executing (often iterative) processes to extract, manipulate, and analyze large volumes of data, and shall perform efforts necessary to output results of those data processes in varying formats, as required. For example, the contractor shall support requests to map a set of part number-centric data to the equivalent National Stock Number (NSN)-centric data and vice-versa. In order to support these requests, the contractor shall perform batch and interactive part to item mapping services, to include extraction, transformation, analysis, and annotation and tagging of data, and shall output results based on customer requirements.

5.10.3. Documentation

The contractor shall prepare and maintain documentation to reflect current PDSS infrastructure and software configurations and shall develop and maintain all documentation necessary to operate, maintain, enhance, utilize, or provide training on any process or system component used in the performance of this contract, and shall deliver such documentation within 15 days after receipt of request from the Government technical program staff. The contractor shall also prepare and maintain documentation describing all processes and toolsets utilized to execute contract requirements. In addition, the contractor shall develop and maintain materials to support user training and documentation necessary to support efforts to achieve compliance with system management and security directives, to include portfolio management, and certification/accreditation of assigned systems.

(Exhibit A: CDRL A001, Technical Report-Study/Services, DI-MISC-80508B; Exhibit A: CDRL A003, Software Transition Plan, DI-IPSC-81429A; Exhibit A: CDRL A004, Computer Program End Item Documentation, DI-IPSC-80590B)

All documentation developed shall undergo a currency review and be validated as current by the contractor monthly. The contractor shall maintain an up-to-date continuity binder detailing business processes utilized to support the scope of this PWS and shall make continuity documentation available to the Government program management staff for review upon request. All documentation developed, to include training materials and documentation necessary to operate, maintain, and enhance any process, system, or toolset utilized shall be maintained by the contractor within Government-owned systems and/or repositories and, if requested by the Government Program Manager, shall be collated by system and provided in hard and/or soft copy to the Government within 10 day of receipt of request.

(Exhibit A: CDRL A001, Technical Report-Study/Services, DI-MISC-80508B)

All documentation shall be approved by the Government Program Manager. Documentation shall be updated regularly to reflect current operating environments. Any deficiencies identified shall be corrected by the contractor within 15 working days of deficiency identification. Documentation shall be prepared and maintained for the following:

- Requirements Documents - all existing product data system infrastructure and/or software shall be formally documented and vetted through a CCB. Government personnel shall be the final signature authority on all CCB actions. The Requirements document must include input from lead personnel supporting all required system development and/or sustainment disciplines, to include infrastructure, software, testing, security, and the functional user community. PDSS infrastructure and/or software will not be deployed until all required documentation is accomplished, unless formal waiver is authorized by the Government.
- Change Requests (CRs) - all changes to existing PDSS infrastructure and/or software shall require a detailed CR document and shall be vetted through a CCB. Government technical program staff shall be the final signature authority on all CCB actions. The CR document must include input from lead personnel supporting all required system development and/or sustainment disciplines, to include infrastructure, software, testing, security, and the functional user community. Changes to PDSS infrastructure and/or software will not be deployed until all required documentation is accomplished unless formal waiver is authorized by the Government.
- Issue Tracking - all issues associated with deployed PDSS infrastructure and/or software shall be input in an issue tracking system such as JIRA. Each issue shall be associated with a CR.
- Process Documentation - all processes executed in support of the requirements of this PWS to include, but not limited to those associated with CCB activities, infrastructure testing and deployment, software testing and deployment, PDF/QA, PCMS/D200F, VIPR, FTS, system backup, 2875 processing, user account creation, daily system sustainment and Continuous Process Improvement (CPI) shall be formally documented.
- Training Documentation - user guides and tip sheets shall be prepared for supported systems, to include the TD-PDM, TOAP, and STOR systems. Delivery of user guides and tip sheets for newly developed or modified systems shall occur prior to system production deployment. Detailed classroom training material shall be prepared and maintained for both SGML and FrameMaker authoring tools, to include instruction materials necessary to facilitate use of Government-owned templates.
- CTOR PDF QA/PCMS Workload Tracker Report - refer to paragraphs 5.10.1.3 of this PWS - This monthly report will include all TOs and increments loaded as well as PCMS files loaded.
- CTOR Load Activity Report - refer to paragraphs 5.10.1.3 of this PWS - This monthly report shall contain a breakdown of loaded TOs/Pages by program and totals for TOs/Pages. Report shall also document counts for rescinded/reinstated records, urgent jobs, and TOMA updates.
- CTOR Load Discrepancy Report - refer to paragraphs 5.10.1.3 of this PWS - This monthly report shall include discrepancies and other issues identified during the CTOR pre-load process and any return files and their disposition needed to correct the file and load into CTOR.
- Resource Allocation (Material) - refer to paragraph 3.1.2 of this PWS - This monthly report shall document each material purchase on the task order and detail what PDSS program and sub-elements are supported by the material purchase.
- Resource Allocation (Labor) - refer to paragraph 6.12.2 of this PWS - This monthly report shall provide a detailed breakout of labor hours by PDSS program, and sub-elements within each program. Breakout shall document hours by FTE including functional area supported, labor rates, and labor category.

- Contractor Roster with Training Status - refer to paragraph 6.12.3 of this PWS - This monthly report shall document the status of every government required training activity by contractor with completed training dates.
- Meetings and Conferences - refer to paragraph 5.10.6 of this PWS - These documents shall track the meetings executed with both internal team and external customers and include action items identified with a specific primary point of contact for investigating and tracking the action item; trip reports shall be provided following any trips performed in support of PDSS.
- Program Management/Reports - refer to paragraph 5.10.4 of this PWS - Monthly reports shall include a monthly program management review (PMR) and monthly status report (MSR).
- Transition Plan - refer to paragraph 6.16 of this PWS - These documents shall be provided upon request of the PMO detailing processes and solutions performed on the PDSS effort.

5.10.4. Project Management/Reports

The contractor must document task, schedule, and expenditure plans in a Program Management Plan (PMP) and submit the plan to the Government Program Manager within 30 days after contract award. The established PMP shall be considered a living document and will routinely be updated to reflect the current status of program activities. Comments to the PMP from the Government shall be incorporated, and the updated PMP shall be resubmitted to the Government for approval within 15 days of receipt of Government comments.

(Exhibit A: CDRL A007, Program Plan, DI-MGMT-81797)

The contractor shall generate monthly progress, status, and management reports outlining progress for the previous month and plans for the upcoming month. Monthly reports shall also outline monthly and cumulative financial expenditures. Financial data presented in reports must be accurate. Financial errors exceeding \$1,000 on any given report are considered critical and will result in performance being deemed unacceptable. Program Management Reviews (PMR) shall be completed, submitted, and briefed on a monthly basis and shall include the contract information, CDRL status, performance metrics, staffing update, financial updates, material spreadsheet, travel spreadsheet, risks/issues and proposed resolutions (as required).

(Exhibit A: CDRL A002, Status Report, DI-MGMT-80368A)

The contractor shall be responsible for documenting the requirements and executing processes necessary to automate and capture required metrics to generate reports related to all task areas. Metrics shall be comprehensive and reported monthly or as requested by the Government Program Manager.

(Exhibit A: CDRL A007, Program Plan, DI-MGMT-81797, Exhibit A: CDRL A002, Status Report, DI-MGMT-80368A)

The contractor shall be responsible for creating and maintaining project management plan utilizing Microsoft Project to track contractor workload, assigned resources, planned release dates and other information as requested by the Government Program Manager.

(Exhibit A: CDRL A007, Program Plan, DI-MGMT-81797)

Projects executed at the direction of the Government Program Manager and Technical and Functional Leads should be performed within the associated Period of Performance (POP), where reasonable. Individual planned project deliveries should not exceed 12 months. If a project is planned over a multi-year schedule, then pieces of that effort should be clearly defined in the Microsoft Project plan of what will be executed within the POP and what will be workload performed outside of the POP and approved by Government Program Manager. Adjustments made to the Microsoft Project plans can be incorporated through the life of the project and POP, in order to maintain the most current expectation of deliverables during the POP and planned for future POPs.

5.10.5. Data Rights

The Government shall have unlimited rights to all noncommercial software developed under this contract. Noncommercial source code for all systems, scripts, software tools, etc., developed under this contract must be thoroughly documented and shall be delivered unencumbered by restrictive markings to the Government at contract completion or within 30 days after receipt of a request from the Government Program Manager.

(Exhibit A: CDRL A003, Software Transition Plan, DI-IPSC-81429A; Exhibit A: CDRL A004, Computer Program End Item Documentation, DI-IPSC-80590B)

Commercial software licenses shall also transfer to the Government at contract completion, or within 30 days after receipt of a request from the Government Program Manager.

(Exhibit A: CDRL A003, Software Transition Plan, DI-IPSC-81429A)

All development efforts shall be approved by the Government Program Manager prior to start of development. All custom-developed noncommercial software shall be properly documented using industry or DAF standard techniques.

(Exhibit A: CDRL A004, Computer Program End Item Documentation, DI-IPSC-80590B)

5.10.6. Support Meetings/Conferences

The contractor shall provide written communications and briefings to Government customers as required and attend meetings and conferences as necessary to support product/technical data and system efforts and strategies. The contractor may be required to prepare and submit documents associated with meetings conducted or attended. Refer to the following attached CDRLs for document requirements.

(Exhibit A: CDRL A005, Conference Agenda, DI-ADMN-81249C; Exhibit A: CDRL A006, Report, Record of Meetings/Minutes, DI-ADMN-81505)

5.10.7. Reserved

5.10.8. Reserved

5.10.9. Reserved

5.10.10. Required Skills

The contractor shall have an extensive working knowledge of the applications resident within the Robins AFB PDSS infrastructure and with equipment and tools utilized in the integration and management of technical data. The contractor shall possess the knowledge base necessary to adequately diagnose and

resolve problems associated with the system infrastructure, and with technical data production, output, and usability. The contractor shall be responsible for ensuring all training and certifications requisite to gaining access to, and performing system support functions on systems attached to, the Robins AFB unclassified and classified networks and the supported technical data infrastructure are obtained and maintained by contract support staff. In addition, the contractor staff shall possess comprehensive skills in the following areas at a minimum:

Required Skills		
.Net Development Suite	Manage Engine ServiceDesk	STIG Viewer
Abby FineReader	Microsoft Office Suite	Storage Area Network (SAN)
Adobe Acrobat Professional	Microsoft SQL Server	TCP/IP Network Protocol
Arbortext Editor Software	Oracle Relational Database Management System	Teamcenter Product Lifecycle Management System
Assured Compliance Assessment Solution (ACAS)	Internet Information Server	Teamcenter Reporting and Analytics
Authorization and Accreditation (A&A) Artifact Authoring	PCMS-GUI	Timestep Software
Cisco Multilayer Director Switches (MDS)	PTC Arbortext Editor	TomCat and Apache Web Services Applications
ConfigOS	Veritas Netbackup	Veritas Storage Appliance
Ansible	Public Key Infrastructure	UltraEdit
Enterprise Mission Assurance Support Service (eMASS)	QlikSense	Uninterruptible Power Supplies
ESX, P2V, and Virtual Center Applications and Servers	Redundant Array of Independent Disks (RAID)	Unix/Linux/Solaris Operating Systems
FrameMaker Desktop Publishing Suite	RWS Contenta/Contenta S1000D Content Management Systems	Various Computer Aided Design (CAD) and Illustration Software Applications
Congruity	RWS LiveContent	Veritas NetBackup
Internet System Security	RWS XML Professional Publisher (XPP)	Visual Basic
Java	Security Content Automation Protocol (SCAP)	Windows Workstations
JBoss	Security Content Automation Protocol (SCAP)	Windows Server Operating Systems
Atlassian JIRA	SGML/HTML/XML	Microsoft Defender
Antivirus Software	Splunk	Service Desk

Contract personnel supporting product data and desktop systems shall be responsible for ensuring compliance with regulatory guidance governing DAF information systems. Personnel assigned to PDSS, desktop, and user support positions are required to be familiar with guidance contained in applicable DAF and DoD guidance documents and will be required to maintain current knowledge of regulatory guidance pertaining to the function supported.

6. GENERAL INFORMATION: PRODUCT DATA SYSTEMS SUPPORT

6.1. Security Requirements

6.1.1. Security Regulations

The contractor shall ensure personnel, information, system, property, facility and, if applicable, international security requirements are met. The contractor shall ensure contractor/subcontractor personnel who perform work on a Government facility comply with the security requirements of the facility. The contractor shall comply with the following security regulations:

- DOD 5200.01- DoD Information Security Program
- DOD 5200.2 - Personnel Security Program
- 32 CFR Part 117 - National Industrial Security Program Operations Manual (NISPOM) Rule
- DAF and Local Directives, Instructions and Other Policy Documents Governing System Security Requirements and Procedures
- DAF and Local Directives, Instructions, and Other Policy Documents Governing Handling, Use, Distribution, and Disposition of Government/Military Technical Data

6.1.2. Operations Security (OPSEC)

The purpose of OPSEC is to reduce the vulnerability of USAF missions to adversary collection and exploitation of critical information. DoDD 5205.02E and DoD Manual 5205.02-M require contractors supporting DoD activities use OPSEC to protect critical information for specified contracts, subcontracts, and information used in the performance of duties in support of Air Force requirements/missions. The contractor shall ensure contractor personnel who perform work on (or in) a Government facility comply with the OPSEC procedures and training requirements of the facility as specified in AFI 10-701.

6.1.3. Communications Security (COMSEC)

The contractor shall use only secure communications methods and/or equipment to transmit or otherwise transfer classified information and/or Controlled Unclassified Information (CUI) in accordance with 32 CFR Part 117, *National Industrial Security Program Operating Manual (NISPOM)*, § 117.7(a). Applicable equipment shall be safeguarded, maintained and operated in accordance with 32 CFR Part 117, *National Industrial Security Program Operating Manual (NISPOM)*, § 117.7(a).

6.1.4. Security Clearance

The contractor shall ensure applicable contractor/subcontractor personnel have security clearance(s) at the Favorable National Agency Check and Inquiries (NACI) level for proper accomplishment of contract/order requirements. The security clearance(s) shall be obtained in accordance with the Department of Defense (DD) Form 254, Department of Defense Contract Security Classification Specification.

The contractor's Facility Security Officer (FSO) will be responsible for processing of the documentation necessary for contractor employees to obtain the appropriate background investigations as specified by DoD 5200.2 for any level of access above unclassified. Each employee requiring a background

investigation for this purpose will comply with instructions furnished by the cognizant security manager and the contractor's FSO.

The contractor must possess a facility clearance at the classification level indicated on the associated DD Form 254, Department of Defense (DoD) Contract Security Classification Specification to be eligible for award. It is anticipated that some task orders will be classified, requiring processing and appropriate protection measures for data classified at levels up to and including SECRET.

The contractor shall obtain security clearances or background checks, as appropriate, for all personnel performing duties in support of the program. Every employee must be a United States citizen and must have favorable background investigation results. All personnel performing duties associated with classified data will possess a SECRET clearance. Access to Government automated information system resources which process sensitive, unclassified data and are considered Controlled Unclassified Information (CUI) will be required. Therefore, as a minimum, NACIs will be conducted for all personnel in accordance with DoD 5200.2.

The contractor shall be capable of meeting security requirements for classified and unclassified task orders for both on-base and off-base tasks. The contractor shall not divulge any information regarding files, data processing activities/functions, user identifications, passwords, or other knowledge that may be gained, to anyone who is not authorized to have access to and does not have a need to know such information. The contractor personnel performing under the program shall abide by all Government directives, rules, procedures, and standards of conduct.

The contractor shall provide a listing of contractor employees' names, position titles and security clearance level to the Government technical program staff prior to employee start date. The contractor shall also provide an updated listing of personnel to the Government technical program staff within five (5) workdays of contract employee status or position changes.

The security clearance(s) shall be obtained in accordance with the DD Form 254, Department of Defense Contract Security Classification Specification.

6.1.4.1. Network Access Foreign Nationals

Not Applicable.

6.1.5. Security Incident or Violation

The contractor shall immediately notify the Government Security Office of any potential or actual security incident or violation.

6.1.6. Access to Government Systems

The Government will provide contractor/subcontractor personnel access to system(s) necessary to perform tasks under the contract/order. The contractor shall ensure contractor/subcontractor personnel who require access to an unclassified or classified Government Automated Information System (AIS) have the respective background check or security investigation conducted. At a minimum, the contractor shall ensure contractor/subcontractor personnel who require access to an unclassified AIS have a National Agency Check with Inquiries (NACI) conducted. Upon completion/termination of the contract/order or transfer/termination of contractor/subcontractor personnel, the contractor shall ensure

the system account(s) are closed. If contractors will be accessing Government Shared Drives or SharePoint sites, users must complete assigned unit initial and annual training.

6.1.7. Access to Robins AFB

The contractor shall ensure contractor/subcontractor personnel who require access to Robins AFB comply with the security requirements of the installation. The contractor shall provide the PCO a list of all contractor/subcontractor personnel requiring access to Robins AFB, the contract/order number and the period of performance. The contractor shall also provide, if possible, the location of the work site and the days/hours during which the contractor/subcontractor personnel will require access to Robins AFB.

6.1.7.1. Contractor Credentials and Vehicle Entry

For contract performance on a military installation, contractor shall comply with local procedures provided by the SSA to obtain credentials and vehicle entry.

6.1.8. Contractor Identification

All contractor management staff and personnel shall clearly be identified as such at all times and from a distance. Contractor identification shall include conversations, mail, email, teleconferences, video teleconferences, faxes, and/or other electronic communication whether with Government personnel, other contractor personnel, or with the public when supporting this contract where their status as contractor employees might not otherwise be apparent or where they might be mistaken for civil service employees. As a minimum, contractor management staff and personnel shall clearly identify themselves using the label “contractor” in email correspondence in accordance with Federal Acquisition Regulation (FAR) 37.114 and AFI 33-119. Contractor badges shall be worn by contract management staff and personnel, will be visible at all times, and shall be worn above the waistline during the individual’s duty hours. Where practicable, contractors/subcontractors occupying collated space with their Government program customer should identify their workspace area with their name and company affiliation.

6.1.8.1. Unescorted Entry Authorizations

To obtain unescorted entry authorizations into controlled or restricted areas, the contractor shall comply with local procedures provided by the SSA. For the purpose of this program, the SSA is the local security organization that provides security support for all Government and contractor personnel performing on the installation. Forms accomplished to obtain such entry must be certified by the SSA to ensure the individual requiring unescorted entrance has a personnel security clearance or a favorable background investigation.

6.1.9. Common Access Card (CAC)

The contractor shall ensure a CAC is obtained by all contractor/subcontractor personnel who require access to DoD computer networks/systems, for DoD facility entry control and/or for physical access to facilities and buildings to perform tasks under the contract/order. The contractor shall provide a list of contractor/subcontractor personnel who require a CAC to the PCO. The Government will provide the contractor instruction on how to utilize Mission Partner Identity, Credential and Access Management (MP-ICAM) and notify the contractor of approval/disapproval of contractor/subcontractor personnel application. Contractor/Subcontractor personnel shall obtain the CAC from the local Real Time Automated Personnel Identification Documentation System (RAPIDS) issuing facility [usually the Military Personnel Flight (MPF)].

The contractor shall immediately report a lost or stolen CAC as directed by local Government policy. The contractor shall notify the PCO of any change to the list of contractor/subcontractor personnel who require a CAC and provide an updated list within five business days. The contractor shall return a CAC within five business days once contractor/subcontractor personnel no longer require computer network/system access and/or facility access. The contractor shall return an expired CAC within five (5) business days after the expiration date. The contractor shall return any and all CACs within five (5) business days after completion/termination of the contract as directed by local Government policy.

6.2. Environmental Management (EMS)

Contractor/Subcontractor personnel who perform work on any Government facility shall comply with the EMS requirements established by the facility. Contractor/Subcontractor personnel may be required to complete EMS training prior to beginning work.

6.3. Agency Affirmative Procurement Programs

In accordance with FAR 23.404, Agency Affirmative Procurement Programs, 100 percent of purchases of Environmental Protection Agency (EPA)-designated products included in the Comprehensive Procurement Guidelines (CPG) list [<http://www.epa.gov/cpg/products.htm>] and/or United States Department of Agriculture (USDA)-designated products included in the bio-based product listings [<http://www.biopreferred.gov>] shall, at a minimum, meet the EPA or USDA standards for recovered materials or bio-based content, respectively, unless an item cannot be acquired competitively within a realistic timeframe, meet appropriate performance standards, and/or be acquired at a reasonable price.

6.4. Safety Requirements

The contractor shall demonstrate capabilities to comply with safety provisions, i.e., technical specifications, technical publications, Federal Occupational Safety and Health Administration (OSHA) standards, and other nationally or locally recognized sources of safety, health, and fire prevention standards and practices. The contractor shall comply with all safety requirements 100 percent of the time.

6.4.1. Mishap Notification/Investigation

The contractor shall report mishaps including damage to DoD property; occupational illness to DoD military or civilian personnel; injury to on- or off-duty DoD military personnel; injury to on-duty DoD civilian personnel; and damage to public or private property or injury or illness to non-DoD personnel caused by Government operations. The contractor shall ensure the Safety Office and the PCO are notified of mishaps. The contractor shall contact the Multi-Functional Team (MFT) (COR, if available, or another MFT member) by telephone within 24 hours of the accident/mishap. The contractor shall report all mishaps/incidents in accordance with the contract. The contractor shall immediately secure the mishap scene and damaged property and impound pertinent maintenance and training records until released by the Government's Safety Office (WR-ALC/SE). Such release shall be accomplished through the CO. The contractor shall cooperate and assist Government personnel in the investigation of the incident and submit an Accident/Incident Report within 24 hours of the accident/incident. The contractor shall require all subcontractors to comply with required safety, health and fire standards.

(Exhibit A: CDRL A009, Accident/Incident Report, DI-SAFT-81563/T)

6.4.2. Safety and Health Plan/Program

The contractor shall develop and maintain a Health and Safety Plan, reference *Attachment 02_Ind Safety and Health_PDSSIII.pdf* in the proposal documents for specific instructions. For task specific procedures identified in *Attachment 02_Ind Safety and Health_PDSSIII.pdf*, the contractor's internal procedures for those tasks must be identifiable within the submitted Health and Safety Plan or shall be provided in an amendment or addendum to the company's Safety Plan and included as part of the Program Management Plan. The contractor will have 40 days after contract award to submit the plan or addendum to address requirements. The 78 ABW Safety Office (78 ABW/SEG) will provide acceptance or non-acceptance to the Government COR.

(Exhibit A: CDRL A008, Environmental, Safety, and Occupational Health (ESOH) Plan, DI-SAFT-82313)

6.5. Inspection of Services

In accordance with the Inspection of Supply/Services clause(s) identified in the contract/order, the Government reserves the right to inspect contractor performance.

6.6. Service Contract Reporting (SCR)

In accordance with DFARS 252.204-7023, *Reporting Requirements for Contracted Services*, the contractor shall provide a Service Contract Report for each contract/order that meets the DoD Contract Reporting criteria and has obligations in excess of \$3,000,000. The contractor shall report the total amount invoiced and contractor labor hours expended on the services performed during the previous fiscal year. Contractor labor hours, including any required Tier 1 subcontractor information, shall be reported in the System for Award Management (SAM). The contractor shall report annually, by October 31, on the services performed during the preceding Government fiscal year. SAM will only enable reporting for entities with contracts that meet the SCR criteria and thresholds. Additional information and user manuals for Government personnel and contractors is available at <https://www.sam.gov>. SCR eligibility, reporting rules and criteria is located on the DoD Procurement Toolbox at <https://dodprocurementtoolbox.com/site-pages/service-contract-reporting-scr>.

6.7. Invoicing/Payment and Receipt/Acceptance

The contractor shall submit/process payment requests and receipt/acceptance documents via WAWF.

6.8. Quality Control Plan/Quality Management (QMS)

The contractor shall ensure the quality of services through a quality and/or inspection system. The contractor shall submit a Quality Control Plan (QCP) addressing detection of quality program problems and defects, identification of root causes for quality related problems/defects, correction of root causes related to detected problems/defects, and follow-up to ensure quality related problems/defects do not recur. The plan shall include the schedule of quality assurance/control inspections planned by the contractor. 100% compliance with the QMS is required.

(CDRL Reference in paragraph 4.2).

6.8.1. Self-Assessment Program (SAP)

The contractor will maintain an internal SAP to evaluate the effectiveness of existing procedures and internal controls, to determine organizational compliance with regulatory requirements and to

complement external inspections and assessments. Additionally, the contractor shall provide support, as required, for contingencies, exercises, and surge operations. The Government Program Manager will notify the contractor as soon as notification of upcoming event becomes known.

6.8.2. Quality Control and Continuous Process Improvement (CPI)

The contractor shall support DAF initiatives and participate in other applicable Quality Control programs and CPI initiatives designed to indicate the appropriate end-state of managed TOs. The contractor shall comply with resulting changes to existing processes/procedures from these programs/initiatives. (AFMCI 90-104)

6.9. Trafficking in Person

The contractor shall comply in accordance with the FAR and applicable supplements and shall be in compliance with all applicable guidance and clauses listed in the contract as it relates to Trafficking in Persons. Additional information about Trafficking in Persons is available at the site for the Department of State's Office to Monitor and Combat Trafficking in Persons. <http://www.state.gov/j/tip>

6.10. Continuation of Mission Essential Services During a Crisis

6.10.1. Designation of Services as Mission Essential

The Functional Commander (FC) or civilian equivalent has determined certain services under this contract/order are mission essential and will continue in the event of a crisis. The designation of services as mission essential will apply to an entire contract but will apply only to services specifically identified as mission essential by the FC or civilian equivalent. The mission essential services under this contract/order include that the services outlined in this PWS.

6.10.2. Mission Essential Contractor Services

Contractor personnel may be required to provide mission essential services during a crisis.

6.10.2.1. Contingency Plan

The contractor shall submit a contingency plan describing procedures for continuing performance of mission essential services during a crisis. The contractor shall identify provisions for ensuring required personnel and resources are available to support continuity of operations for up to 30 calendar days or until normal operations can be resumed. The contractor shall submit any revision to the plan for approval prior to implementation. The contractor shall activate the plan only if authorized by the PCO acting under the direction of the FC or civilian equivalent. The plan shall address the following elements:

- Identify provisions for the acquisition of necessary personnel and resources, if necessary for continuity of operations for up to 30 days or until normal operations resume.
- Challenges associated with maintaining contractor essential services during an extended event.
- The time lapse with the initiation of the acquisition of necessary personnel and resources and their actual availability on site.
- The components, processes and requirements for the identification, training, and preparedness for personnel who are capable of relocating to alternate facilities or performing work from home.

- Any established alert and notification procedures for mobilizing identified “essential contractor service” personnel.
- The approach for communicating expectations to contractor employees regarding their roles and responsibilities during a crisis.

6.11. Nondisclosure Requirements

Contractor personnel will be exposed to information covered under the Privacy Act of 1974 and shall comply with all applicable safeguarding and handling requirements associated with Privacy Act information. During the performance of the requirements, contractor personnel may be required to access or come into contact with proprietary information or sensitive Government information, including financial account records or source sensitive information. Contractor personnel shall be required to execute Nondisclosure Agreements and shall agree not use such information for any purpose other than in the performance of the requirements outlined within this PWS. Additionally, contractor personnel shall be required to sign user agreements associated with system access which prohibit use of data accessed for any purpose other those outlined within applicable contract documents. The contractor shall not release, remove, or replicate in whole or in part, any system documentation, data, or reports generated by or through use of any Government systems for use other than to support requirements outlined within this PWS. All requests for information shall be forwarded to the Government Program Manager or Contracting Officer issuing this contract.

The following shall apply to contractor personnel performing in support of requirements:

- The performance chain for contract personnel assigned to support requirements will not fall under their company’s Business Development or Proposal Management organizational structures.
- The physical work location for contractor personnel assigned to support requirements will be separate and isolated from their company’s Business Development work area.
- Contractor personnel assigned to support requirements will not divulge any competition sensitive Government information that they have access to as a result of their positions. Furthermore, contractor personnel assigned to support the requirements outlined within this PWS will not work as representatives for their company for other contract solicitations that are either issued by the Government program office managing these requirements or for which the contract program management organization is responsible for source selection.

6.12. Personnel Reporting

The contractor shall report ALL contractor labor hours (including subcontractor labor hours) required for performance of services provided under this contract for Product Data Systems, Data Management, and Migration Support via a secure data collection site. The contractor is required to completely fill in all required data fields at <http://www.ecmra.mil>. Reporting inputs will be for the labor executed during the period of performance for each Government fiscal year (FY), which runs 1 October through 30 September. While inputs may be reported any time during the FY, all data shall be reported no later than 31 October of each calendar year. Contractors may direct questions to the CMRA help desk.

The contractor shall supply a Labor Resource Allocation report, monthly, detailing labor hours by major program and subcategories for monthly labor reporting. This report shall show a detailed view of labor required to support each element of the Robins AFB PDSS major programs.

The contractor shall report personnel data updates monthly to record such changes as on-boarding, out-processing, team changes, labor category changes, actual and composite labor rates, status of required training, etc.

6.13. Uses and Safeguarding of Information

Information from the secure web site is considered to be proprietary in nature when the contract number and contractor identity are associated with the direct labor hours and direct labor dollars. At no time will any data be released to the public with the contractor name and contract number associated with the data.

6.14. User Manuals

Data for DAF service requirements must be input at the DAF CMRA link. However, user manuals for Government personnel and contractors are available at the Army CMRA link at <http://www.ecmra.mil>.

6.15. Non-Personal Services

The Government will neither supervise contractor employees nor control the method by which the contractor performs the required tasks. Under no circumstances shall the Government unilaterally assign tasks to, or prepare work schedules for, individual contract employees. It shall be the responsibility of the contractor to manage its employees and to guard against any actions that are of the nature of personal services or give the perception of personal services. Pursuant to Federal Acquisition Regulation (FAR) 37.104, personal services are prohibited unless authorized by statute. If the contractor feels that any actions constitute, or are perceived to constitute personal services, it shall be the contractor's responsibility to notify the Government Program Manager immediately. These services shall not be used to perform work of a policy/decision making or management nature, i.e., inherently Government functions. All decisions relative to programs supported by the contractor shall be the sole responsibility of the Government.

6.16. Transition/Phase Out

It is essential to the Government that services required are performed without significant interruption. At the conclusion of any performance period, including option periods, the services provided under this PWS may be awarded to a successor contractor. In such an event, the incumbent contractor shall be required to assist and cooperate in transition activities to ensure a well-planned, orderly, and efficient transition. To ensure services are continued with minimal interruption should a transition be required, the incumbent contractor shall assist the Government in efforts associated with transition planning and shall document processes for the transition of powers, duties, documentation, activities, and functions associated with tasks and tools covered under this PWS. This documentation shall be subject to review and approval by the Government. The Government estimates that complete transition of all performance areas may take 180 days or more. Immediately upon notification that a transition will occur, the incumbent contractor shall identify employees to serve on an integrated transition team consisting of Government, incumbent contractor, and successor contractor members. The incumbent contractor shall, as a minimum, appoint program management and project and functional area lead personnel to the integrated transition team, and shall augment the team as needed with personnel knowledgeable of specific processes.

6.17. Government Points of Contact

Joseph Tompkins, AFLCMC/LZPPS
 Phone: (478) 222-4698
 E-Mail: joseph.tompkins.12@us.af.mil

Robyn Gallagher, AFLCMC/LZPPS
 Phone: (478) 222-3187
 E-Mail: robyn.gallagher.2@us.af.mil

Rachel Sims, AFLCMC/LZPPS
 Phone: (478) 222-3186
 E-Mail: rachel.sims.2@us.af.mil

7. APPENDICES**7.1. Appendix A. Reference(s)**

Publication / Date	Title of Publication
32 CFR Part 117	National Industrial Security Program Operating Manual (NISPOM).
DoDD 4715.1E	Environment, Safety, and Occupational Health (ESOH)
DoDD) 5205.02E	DoD Operations Security (OPSEC) Program
DoD Manual 5205.02-M	DoD Operations Security (OPSEC) Program Manual
Public Law 91-596 as amended by Public Law 101-552, 29 Dec 1970 as amended through January 1, 2004	Occupational Safety and Health Act of 1970
MIL-DTL-38769G(1) 27 Mar 2020	Detail Specification Manuals, Technical: Work Unit Code
MIL-DTL-38804F 3 Jun 2021	Detail Specification Manuals, Technical - Time Compliance Technical Orders (TCTO) and TCTO Supplements
MIL-DTL-38807D(3) 6 Jun 2024	Detail Specification Manuals, Technical - Illustrated Parts Breakdown
MIL-DTL-5096J 13 May 2022	Detail Specification Manuals, Technical – Inspection and Maintenance Requirements; Inspection Work Cards; and Checklists; Preparation of
MIL-DTL-5288K(2) 12 Mar 2024	Detail Specification Manuals, Technical - Aircraft Cargo Loading and Off-Loading
MIL-DTL-7700K 11 Aug 2022	Detail Specification Manuals, Technical - Flight Manual Program Publications
MIL-DTL-8031H(2) 22 Apr 2020	Detail Specification Manuals, Technical - List of Applicable Publications (LOAP) - Preparation
MIL-DTL-87158D(2) 19 Dec 2022	Detail Specification Manuals, Technical - Aircraft Battle Damage Assessment and Repair
MIL-DTL-87268D(1) 19 Jul 2019	Detail Specification Interactive Electronic Technical Manuals General Content, Style, Format, and User-Interaction Requirements
MIL-DTL-87269D(1) 30 Jan 2019	Detail Specification Database, Revisable Interactive Electronic Technical Manuals, for the Support of

Publication / Date	Title of Publication
MIL-DTL-9854F 11 Oct 2024	Detail Specification Manuals, Technical - Structural Repair (Aircraft)
MIL-DTL-9977N(2) 27 Jun 2024	Detail Specification Munitions/Nuclear Weapons Basic Information and Loading Technical Manuals and Checklists
MIL-HDBK-28001 30 Jun 1995 Notice 1 / 22 Dec 2000 Notice 2 / 4 Apr 2010	Department of Defense Application of MIL-PRF-28001 Using Standard Generalized Markup Language (SGML)
MIL-PRF-28000B 30 Sep 1999 Notice 1 / 22 Apr 2010	Digital Representation for Communication of Product Data: IGES Application Subnets and IGES Application Protocols
MIL-PRF-28001C 2 May 1997 Notice 1 / 22 Apr 2010	Markup Requirements and Generic Style Specification for Exchange of Text and its Preparation
MIL-PRF-28002C 30 Sep 1997 Notice 1 / 22 Apr 2010	Raster Graphics Representation in Binary Format
MIL-PRF-28003B 30 Apr 2000 Notice 1 / 22 Apr 2010	Digital Representation for Communications of Illustration Data
MIL-STD-1808C(2) 16 Apr 2020	Department of Defense Interface Standard System Subsystem Sub-Subsystem Numbering
MIL-STD-3048C 15 May 2023	Department of Defense Standard Practice Air Force Business Rules for the Implementation of S1000D
MIL-STD-38784C 4 Sep 2024	Department of Defense Standard Practice General Style and Format Requirements for Technical Manuals
TO 00-5-3 29 Nov 2023	Air Force Technical Order Life Cycle Management
TO 00-5-15 28 Jul 2023	Air Force Time Compliance Technical Order Process
TO 00-5-18 15 Dec 2022	AF Technical Order Numbering System
DoDI 5200.01 Change 2 / 1 Oct 2020	DoD Information Security Program and Protection Of Sensitive Compartmented Information (SCI)
DoDI 5200.02 Change 3 / 24 Sep 2020	DoD Personnel Security Program (PSP)
DoDI 5220.22 Change 3 / 10 Dec 2021	National Industrial Security Program (NISP)
AFI 17-130 13 Feb 2020	Cybersecurity Program Management
AFMAN 17-1301 12 Feb 2020	Computer Security (COMPUSEC)

Publication / Date	Title of Publication
SAE EIA-649C 7 Feb 2019	Configuration Management Standard
SAE EIA-649-1A 10 Aug 2020	Configuration Management Requirement for Defense Contracts
ASD S1000D Rev 4.2	International Specification for the Procurement and Production Technical Publications
NIST SP 800-53A Rev 5 / Jan 2022	Assessing Security and Privacy Controls in Federal Information Systems and Organizations
NIST SP 800-46 Rev 2 / 29 Jul 2016	Guide to Enterprise Telework, Remote Access, and Bring Your Own Device (BYOD) Security
DODI 8500.01 Change 1 / 7 Oct 2019	Cybersecurity
DOD 8510.01 19 Jul 2022	Risk Management (RMF) for DoD Systems
DOD 8570.01-M Change 4 / 10 Nov 2015	Information Assurance Workforce Improvement Program
DFARS 252.204-7008 29 Sep 2022	Compliance with Safeguarding Covered Defense Information Controls
DFARS 252.204-7012 29 Sep 2022	Safeguarding Covered Defense Information and Cyber Incident Reporting
AFI 17-101 6 Feb 2020	Risk Management Framework (RMF) for Air Force Information Technology
AFI 23-109 13 JUL 2023	Applications, Programs and Indentures (D200F)

7.2. Appendix B. Product Data Systems Hardware and Application Descriptions

339 Interim Tracking and Reporting System (ITRS) - Web-based application developed using asp.net with underlying SQL database utilized as a repository for Defense Logistics Agency (DLA) Forms 339, Request for Engineering Support Documents. The system is utilized locally at Robins only to track billing data and to provide monthly status and accounting reports on the processing of DLA Forms 339 at Robins AFB. 339 ITRS tracks additional data that other Form 339 support tools do not and is specifically applicable to processes utilized at Robins. There are no interfaces between 339 ITRS and other 339 support systems. As an automated tool, 339 ITRS is designed.

Adobe Creative Cloud - A collection of Adobe software applications, including Photoshop, Illustrator, Premiere Pro, InDesign, and more, allowing them to create various digital content like graphic designs, videos, websites, and photo edits.

Adobe FrameMaker - COTS desktop publishing editor used for authoring both structured and unstructured technical documents.

Atlassian suite, JIRA Software and Confluence - A web-based application utilized to capture, organize, plan and execute tasks related to developmental and operational level teams. This web-based application allows requirements management teams to document and monitor work orders more

effectively. With a built-in dashboard, JIRA provides the capability to monitor and track end to end progress of a project.

Automated Data Cleanup (ADC) - Desktop application written in VB.NET that does not use any external databases or system. The ADC systematically executes utilities based on established business rules previously executed manually. It provides features that result in highlighting of cells of suspect records to facilitate further data analysis and investigation, and formats records to remove unnecessary formatting.

Centralized Technical Order Repository (CTOR) - An electronic repository of active and historical TO documents utilized to satisfy view- and print-on-demand processes.

Computer Aided Design/Computer Aided Manufacturing (CAD/CAM) Repository - A repository for the management and storage of native and neutral CAD/CAM source data files.

Congruity - Web services used to track and monitor access to commercial subscription data obtained via separate contract awarded and managed by the DAF.

DAF TO Archive Repository - A repository utilized for archive and retention (for 30 years after rescind date) of electronic copies of DAF TO and TCTO documents and associated increments (i.e., changes, supplements), which supports weapon system programs throughout the DAF.

ELAN ELIP - COTS product used to extract graphic images from PDF files.

ELAN JM - COTS product used to OCR raster images and convert to PDF files.

ELAN TMC - COTS product used for converting paper-based documents to I/PDF documents.

Electronic Technical Order Library (ETOL) - A client interface for CTOR utilized by Robins AFB depot maintenance personnel that replaced paper flight line and backshop maintenance TO libraries, which facilitates printing of, and automatically applies date/time stamp on, TO extracts.

Heavy Clean-up (HC) Tool - Desktop application written in VB.Net with an underlying SQLite database. Also provides viewing mechanism for PCMS .out files, provides querying capabilities utilized to identify data discrepancies, and provides data manipulation and export functionality. Utilized by Equipment Specialist and other technical data support personnel to perform data cleanup functions in preparation for migration of data to DAF-level enterprise systems.

Industrial Plant Equipment (IPE) - IPE manual repository, which contains PDF versions of manuals associated with the operation and maintenance of equipment utilized in the industrial maintenance environment.

Integrated Data Information Manager (IDIM) - IDIM provides integrated workflow processes and configuration management capabilities for linking engineering drawings to contracts, CDRLs, weapon systems, data requests, and data deliveries. The system capabilities support management of tasks and documents required by DAF and DoD technical data acquisition and management policy/guidance.

Integrated Product Data Support Center (IPDSC) - The IPDSC provides infrastructure and software to support product data conversion, development, and sustainment processes.

Interactive Electronic Technical Manual (IETM) Applications - Many of the weapon system program offices have, or are currently, transitioning technical data necessary for operation and sustainment of weapon systems into IETMs. Typically, IETM implementations require administration and sustainment of server system infrastructure, software associated with development, sustainment, transformation, presentation and distribution of technical content data and administration of system user accounts, roles, and privileges, at a minimum. Technical support activities for development support and utilization of IETM implementations at Robins AFB is required.

Maintenance Facility Drawing Repository - A collection of facility drawings occupied/maintained by the industrial maintenance organizations.

Master History Folder (MHF) - A repository of supporting data and work product utilized in the processing of Forms 339, 761, and 762 (AMC/AMSC Screening Analysis Worksheets).

Parts Configuration Management System (PCMS) - A repository of part indenture configuration information (considered Master Bill of Material data) captured from over 19K Illustrated Parts Breakdown TOs.

PCMS Graphical User Interface (GUI) - Desktop application written in VB.Net with an underlying SQL Express 2005 client database. Serves as viewing mechanism for PCMS .out or Maintenance Parts List (MPL) files and allows users to view, manipulate and export data.

Process Order (PO) Repository - A collection of documents outlining local processes which augment the TO documents utilized for depot maintenance processes.

Product Data Management (PDM) Gateway - A web-based user interface which provides access to data within the CTOR, CCR, PCMS and commercial, subscription-based data repositories.

PTC Arbortext Editor - COTS editor used for authoring SGML and/or XML-based technical documents. DAF Document Type Definition (DTD) files are utilized in conjunction with this software.

Public Sales Office (PSO) Tracking System - A web-based application utilized to track actions associated with requests for, and release of product data to the public, the cost associated with satisfaction of those requests, and the payment of those cost by requestors.

QlikSense - COTS comprehensive business intelligence (BI) platform to provide management officials the ability to consolidate data from multiple sources into a single view, explore/evaluate performance, explore data associations, visualize data using graphics, and interact with applications, dashboards, & analytics.

RWS Contenta - COTS Oracle-based content management system. Uses a customized, Government-owned 'bridge' (eBuild Producer) to interact with XPP for SGML composition. Also has an interface for development and sustainment of technical data utilizing the PTC Arbortext Editor and Adobe FrameMaker products, an interface for development, management and sustainment of XML data

developed to the S1000D data specification (Contenta S1000D), and an interface for publishing of S1000D specification based data into IETM output (via Live Content).

RWS Contenta S1000D - SDL Contenta S1000D is a COTS Common Source Data Base (CSDB) that supports content authored to the S1000D specification. Based on the Contenta S1000D content management system, it provides automation needed to develop, manage, sustain and deliver S1000D content. It is used by authors, editors, program and publication managers, and other technical data management personnel for the creation, management and delivery of technical information.

RWS XPP - XML Professional Publisher is a COTS product used for the composition of SGML. Government-owned style libraries based on DAF Military Specifications determine the resultant output (I/PDF) format. Omnimark, Perl, and XyPerl scripts are used in conjunction with this software.

RWS LiveContent - RWS Live Content is a COTS XML product utilized for intelligent retrieval and data mining and provides mechanism supporting creation of Interactive Electronic Technical Publication presentations from XML source data.

Siemens Teamcenter - Teamcenter is a COTS PLM System which provides capabilities to support systems engineering and requirements management, portfolio/program/project management, manufacturing process management, lifecycle visualization, reporting and analytics, and enterprise knowledge management. This product is utilized to facilitate product data sustainment processes and to establish/test/provide capabilities necessary to transition data to DAF enterprise-level systems.

SGML Pre-Processing Tool - A GOTS tool used by TOAP to analyze and automate correction of and/or report on common tagging problems in SGML source data.

Teamcenter PDM Utility - An integrated set of applications designed and developed in Visual Basic, Java and C++ to load meta-data as well as rendered PDF files contained into the CTOR application. The loader is designed and implemented to support infrequent bulk loading of high volume data (during technology refresh or rolling out major feature enhancements) as well as ongoing, routine loading of small amounts of data in support of sustainment. These bulk loading & sustainment loading have distinctly different sets of business rules that are continually evolving. In addition, this utility should be able to support on-demand configuration management & reporting capabilities such as bulk extract, multiple types purge and version roll backs.

Teamcenter Reports and Analytics (TCRA) - The TCRA product is based on the tight integration of the eQube-BI COTS product with the Teamcenter PLM. This product is utilized to provide visibility in the form of key performance indicators (KPIs), Dashboards, detailed reports, and ad-hoc analysis. It provides the capability necessary to form "single view of data" across multiple data sources (to include sources external to Teamcenter). It is built on SOA-compliant open architecture that is highly scalable, and enables rapid collating and leveraging of data from disparate business systems in the Product Data Systems environment.

Technical Manual Information System (TMIS) - GOTS web application used as an archive for historic AFTO 22/252 TO publication change requests for Hill and Robins AFB.

Technical Order Authoring and Publishing (TOAP) - TOAP is the mandatory use, organic TO authoring and publishing component of the DAF Standard TO Management System. TOAP provides the

DAF enterprise capability to manage, author, and publish TO publication source content data and enforces authoring and publishing specification compliance.

Secure Technical Order Repository (STOR) - Secure Technical Order Repository (STOR) is a collection of software tools that provide capabilities for classified data. To include: -Storage, data management, version control, and configuration management capabilities for electronic product data - Repository for electronic TOs - Includes archive repository for USAF classified TOs -Supports view-on-demand (VOD) and print-on-demand (POD) processes -Content Management and Workflow Tools -TO Editing and Publishing Tools -Complete TOs in SGML with all changes created within single SGML instance -Desktop Publishing (FrameMaker) files -S1000D data management and publishing.

UltraEdit - COTS text editor used for a variety of purposes, including technical data authoring and editing, software development, and web development.

8. EXHIBITS

8.1. Exhibit Reference(s):

Exhibit	Title
Exhibit A	CLIN 0004 CDRL(S) A001 – A009